

A Derivative-Free Trust-Funnel Method for Relaxable Simulation-Based Inequality Constraints

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Working draft — M1–M4 complete; smooth measure; self-contained convergence
(TD1+TD2 resolved)

Abstract

We study derivative-free optimization of a simulation-based objective subject to *relaxable*, *quantifiable* simulation-based inequality constraints $c(x) \leq 0$ — constraints that return a meaningful violation magnitude at infeasible points. We propose a trust-funnel (*not* penalty, *not* feasible-iterate) method that builds fully-linear models of both the objective and the constraints on every iteration, and we target convergence to a KKT stationary point under the Mangasarian–Fromovitz constraint qualification. This document fixes the problem class, assumptions, and notation (M1); states the algorithm (M2); develops the convergence analysis (M3); and reports an adversarial self-review with a code-bridge appendix (M4). Using the *smooth* infeasibility measure $\theta = \frac{1}{2} \| [c]_+ \|^2$, we prove that every limit point is feasible ($\theta_k \rightarrow 0$, Theorem 1) and that, under MFCQ, every feasible limit point satisfies the KKT conditions ($\lim_k \pi_k^f = 0$, Theorem 2). A central conceptual point is that building fully-linear models on every iteration is *necessary but not sufficient*: a criticality step and the funnel are the two further ingredients the proof requires. The analysis is *self-contained* — the standard single-region trust-region results it uses are reproduced from first principles and applied to both the objective and the infeasibility merit — so the only inputs are the stated assumptions.

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1 Problem, assumptions, and notation

1.1 The problem class

We study the inequality-constrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c(x) \leq 0, \quad (1)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $c = (c_1, \dots, c_p) : \mathbb{R}^n \rightarrow \mathbb{R}^p$. Both f and c are available only through a *blackbox* / *simulation* that returns their values at a queried point x ; no derivatives are available and finite differencing is assumed unreliable or too expensive. We write the feasible set as

$$\mathcal{F} = \{x \in \mathbb{R}^n : c(x) \leq 0\}. \quad (2)$$

The defining structural assumption — and the source of this project’s novelty — concerns *what the simulation returns at infeasible points*.

Assumption 1 (Relaxable, quantifiable, simulation constraints). Each constraint c_i is *relaxable* (the simulation can be evaluated at points with $c_i(x) > 0$, and such evaluations are permitted by the algorithm), *quantifiable* (the returned value $c_i(x)$ measures the *degree* of satisfaction/violation, not merely a yes/no feasibility flag), and *simulation-based* (revealed only by running the blackbox). This is the QRSK class of [Le Digabel and Wild \(2015\)](#).

Relaxability is exactly what *permits* the normal step and the funnel mechanism of Section 5 to evaluate models built from infeasible sample points; it is the property that fails for the “unrelaxable / feasible-iterate” methods discussed in Section 6.

1.2 Smoothness and the working region

Assumption 2 (Smoothness). f and every c_i are continuously differentiable on $\mathbb{R}^n$, and their gradients $\nabla f, \nabla c_i$ are Lipschitz continuous on an open set containing the level set $\mathcal{L}$ defined below. (These properties are used only in the analysis; the *algorithm* never accesses a derivative.)

Assumption 3 (Bounded level set / working region). There is a point x_0 and a constant such that the region

$$\mathcal{L} = \{x \in \mathbb{R}^n : f(x) \leq f(x_0), \theta(x) \leq \theta_0^{\max}\}$$

swept by the iterates (with θ and $\theta_0^{\max}$ from Sections 3–5) is bounded. Consequently f, c and their gradients are bounded on $\mathcal{L}$.

Assumption 4 (Feasible, reachable start). A point x_0 with $c(x_0) \leq 0$ is given (so $\mathcal{F} \neq \emptyset$).

Assumption 4 is a deliberate scoping decision. Because a feasible start is available, we need *not* prove global feasibility recovery from an arbitrary infeasible start, and we drop the usual “converge to a stationary point of the infeasibility measure” fallback dichotomy. The method may still take *relaxable* excursions into $\{c(x) \not\leq 0\}$ between iterates; the funnel of Section 5 drives the infeasibility back to zero. It also lets the funnel be initialized cleanly (e.g. $\theta_0^{\max}$ from the first tolerated excursion).

1.3 Standing notation

We use $\|\cdot\|$ for the Euclidean norm unless a subscript indicates otherwise, $B(x, \Delta) = \{y : \|y - x\| \leq \Delta\}$ for the closed trust region, and $[a]_+ = \max(0, a)$ componentwise. At iteration k the incumbent is x_k with trust-region radius $\Delta_k > 0$. The names of the algorithmic constants are chosen to match the existing MATLAB scaffold `DFO_quad_TR/Algorithm_11_1.m` and `Algorithm_10_2.m`, so the later implementation is a direct translation:

$$\Delta_{\max}, \eta_0, \eta_1, \gamma \in (0, 1), \gamma_{\text{inc}} > 1, \epsilon_c, \mu, \beta, \omega \in (0, 1), r, \Delta_{\min}.$$

2 Models: fully linear for objective *and* constraints

At x_k we build a model of the objective, m_f^k , and a model of each constraint, $m_{c,i}^k$, valid on $B(x_k, \Delta_k)$. We adopt the standard DFO convention that models *interpolate at the incumbent*: $m_f^k(x_k) = f(x_k)$ and $m_{c,i}^k(x_k) = c_i(x_k)$ (the incumbent is always a sample point), so in particular $\theta_k^m(x_k) = \theta(x_k) = \theta_k$; this makes each predicted decrease $m(x_k) - m(x_k + s)$ a true model decrease from the incumbent. We denote the model gradients and the model Jacobian by

$$g_k = \nabla m_f^k(x_k) \in \mathbb{R}^n, \quad J_k = [\nabla m_{c,1}^k(x_k), \dots, \nabla m_{c,p}^k(x_k)]^\top \in \mathbb{R}^{p \times n}.$$

These are exactly the quantities returned by `pounders/py/formquad.py` for a vector-valued black-box $F = (f, c_1, \dots, c_p)$, together with its `valid` flag (the computational witness of the fully-linear property below).

Definition 1 (Fully-linear model, Conn et al., 2009, Def. 6.1–6.2). A model m of a function $h \in \{f, c_1, \dots, c_p\}$ is *fully linear* on $B(x_k, \Delta_k)$ if there exist constants $\kappa_{\text{ef}}, \kappa_{\text{eg}} \geq 0$, *independent of* k , such that for all $y \in B(x_k, \Delta_k)$,

$$|h(y) - m(y)| \leq \kappa_{\text{ef}} \Delta_k^2, \tag{3}$$

$$\|\nabla h(y) - \nabla m(y)\| \leq \kappa_{\text{eg}} \Delta_k. \tag{4}$$

Assumption 5 (Uniform fully-linearity). The models m_f^k and $m_{c,i}^k$ ($i = 1, \dots, p$) are fully linear on $B(x_k, \Delta_k)$ at *every* iteration, with common constants $\kappa_{\text{ef}}, \kappa_{\text{eg}}$ that do not depend on k . Uniformity is delivered by Λ -poised geometry management (Conn et al., 2009, Ch. 6; Algorithm_6_4.m, Model_Improvement_Step.m), and full-linearity at *every* iteration by running a model-improvement step whenever the incumbent set is not certifiably Λ -poised on $B(x_k, \Delta_k)$ (the standard DFO convention that lets the trust-region lemmas of Section 8.2 apply uniformly). Consequently θ also has a fully-linear model with constants $\kappa_{\text{ef}}^\theta, \kappa_{\text{eg}}^\theta$ (Lemma 1).

We stress the theme this note exists to make precise: *building fully-linear models of f and c on every iteration is necessary but not, by itself, sufficient* for convergence. Two further ingredients are required — a criticality step that ties Δ_k to the criticality measures (Section 4), and a funnel that controls infeasibility (Section 5) — and the analysis of milestone M3 will pin down exactly where each is used.

3 Infeasibility measure

We measure constraint violation by the *smooth* squared measure

$$\theta(x) = \frac{1}{2} \| [c(x)]_+ \|_2^2 = \frac{1}{2} \sum_{i=1}^p [c_i(x)]_+^2, \quad (5)$$

so $\theta(x) = 0 \iff x \in \mathcal{F}$. We deliberately choose the squared form over the unsquared $\|[c]_+\|$: since $t \mapsto [t]_+^2$ is C^1 with derivative $2[t]_+$, (5) is *continuously differentiable* (Assumption 2) with

$$\nabla \theta(x) = \sum_{i=1}^p [c_i(x)]_+ \nabla c_i(x) = J(x)^\top [c(x)]_+, \quad (6)$$

where $J(x) = [\nabla c_i(x)]^\top$ is the constraint Jacobian. This removes the active/violated-boundary nonsmoothness of $\|[c]_+\|$ and makes the feasibility analysis a *smooth* trust-region minimization of θ ; it also matches the measure $\frac{1}{2} \|c\|^2$ of Gould and Toint (2010); Sampaio and Toint (2015), from which we reproduce the requisite trust-region machinery (Section 8.2). Note $\nabla \theta$ is Lipschitz on $\mathcal{L}$ (composition of Lipschitz maps), and θ inherits fully-linear models from those of c (Lemma 1); it is *not* a penalty — infeasibility is controlled by the funnel, never folded into the objective.

The model counterpart of θ , built from the fully-linear constraint models $m_c^k = (m_{c,1}^k, \dots, m_{c,p}^k)$, is

$$\theta_k^m(x) = \frac{1}{2} \| [m_c^k(x)]_+ \|_2^2, \quad \nabla \theta_k^m(x_k) = J_k^\top [m_c^k(x_k)]_+, \quad (7)$$

and its linearization at x_k , $\ell_k(s) = \frac{1}{2} \| [c(x_k) + J_k s]_+ \|^2$, is what the normal step reduces.

4 Two criticality measures

The method monitors two nonnegative quantities that vanish exactly at the two kinds of stationarity we care about.

Feasibility criticality χ_k^c . Because θ is now smooth, its first-order criticality is simply the model gradient norm

$$\chi_k^c = \|\nabla \theta_k^m(x_k)\| = \|J_k^\top [m_c^k(x_k)]_+\|, \quad (8)$$

the standard measure for unconstrained minimization of θ_k^m ; $\chi_k^c = 0$ means x_k is a stationary point of the model infeasibility. (This is the model analog of $\|\nabla \theta\| = \|J^\top [c]_+\|$, the Gould–Toint feasibility measure.)

Optimality / KKT criticality π_k^f . This measures first-order KKT stationarity for (1) using the *model* data (g_k, J_k) and the linearized active constraints. With the model active set $\mathcal{A}_k(\varepsilon) = \{i : c_i(x_k) \geq -\varepsilon\}$, write $J_k^{\mathcal{A}}$ for the rows of J_k indexed by $\mathcal{A}_k$. A convenient projected-gradient form is

$$\pi_k^f = \left| \min_{\|d\| \leq 1, J_k^{\mathcal{A}} d \leq 0} \langle g_k, d \rangle \right|, \quad (9)$$

the largest first-order decrease of the model objective achievable along a direction that keeps the linearized active constraints from being violated. Equivalently, under a constraint qualification, π_k^f is a residual of the KKT system

$$g_k + J_k^\top \lambda = 0, \quad \lambda \geq 0, \quad \lambda_i c_i(x_k) = 0, \quad i = 1, \dots, p, \quad (10)$$

i.e. $\pi_k^f = 0$ certifies that (x_k, λ) solves the *model* KKT conditions. The constraint qualification we invoke to pass from model KKT to true KKT and to bound the multipliers λ is the following.

Assumption 6 (MFCQ at limit points). At every limit point x_* of the sequence of incumbents, the Mangasarian–Fromovitz constraint qualification holds: the gradients $\{\nabla c_i(x_*)\}_{i \in \mathcal{A}(x_*)}$ of the active constraints ($\mathcal{A}(x_*) = \{i : c_i(x_*) = 0\}$) admit a direction d with $\langle \nabla c_i(x_*), d \rangle < 0$ for all $i \in \mathcal{A}(x_*)$.

MFCQ is equivalent to boundedness of the set of Lagrange multipliers and to consistency of the linearized active constraints; both are used in M3 (the normal-step decrease lemma L2 needs linearized consistency, and the passage to a KKT limit point in T2 needs bounded multipliers).

A point x_* is our target *constrained (KKT) stationary point* if $\theta(x_*) = 0$ and there exists $\lambda \geq 0$ with $\nabla f(x_*) + \sum_i \lambda_i \nabla c_i(x_*) = 0$ and $\lambda_i c_i(x_*) = 0$.

5 The trust-funnel mechanism for inequalities

Rather than penalizing infeasibility, we cap it. The method maintains a *funnel bound*

$$\theta_k^{\max} > 0, \quad \text{maintained monotone non-increasing: } \theta_{k+1}^{\max} \leq \theta_k^{\max}, \quad (11)$$

and only accepts iterates that satisfy $\theta_{k+1} \leq \theta_{k+1}^{\max}$. Each iteration is classified (as in Gould and Toint, 2010 and its DFO descendant Sampaio and Toint, 2015) as either

- an *f-iteration*, when the (model-predicted) objective decrease is the dominant achievement and feasibility is comfortably within the funnel; the funnel bound is left unchanged and x_k moves to reduce f ; or
- a *c-iteration*, when reducing infeasibility is the priority; a normal step reduces the linearized violation and the funnel bound is tightened, driving $\theta_k \rightarrow 0$.

The switch between the two is governed by comparing θ_k against a forcing function of the optimality criticality π_k^f (a normal step is taken only when θ_k is large relative to π_k^f), exactly the trust-funnel switching condition, specialized here to the inequality infeasibility measure (5). This single-cap mechanism is the *funnel*; a *filter* (a Pareto set of (θ, f) pairs) is an alternative acceptance rule with the same purpose. We default to the funnel for its cleaner monotonicity proof and note the filter as a variant.

The precise step computation (normal step n_k , tangential step t_k , $s_k = n_k + t_k \in B(x_k, \Delta_k)$), the ratio/acceptance tests, the funnel and Δ_k updates, and the criticality step are specified in milestone M2; their convergence analysis (lemmas L1–L5, theorems T1–T2) is milestone M3.

6 Positioning: why this is a genuine gap

The setting of Section 1 — *model-based* DFO with fully-linear models of the objective *and* of *relaxable inequality* constraints, aiming at KKT stationarity under MFCQ — sits in a gap left open by the closest lines of work. We locate each precedent precisely and identify what it does not cover. Every claim below is stated against the source PDF in [articles/](#).

Trust-funnel lineage (derivative-based). [Gould and Toint \(2010\)](#) introduced the trust-funnel method: equality-constrained nonlinear programming “without a penalty function or a filter,” using a normal/tangential step decomposition and a monotone infeasibility cap (the funnel). We adopt this normal/tangential + funnel machinery (Section 5) but replace derivatives with fully-linear models and target inequalities.

Derivative-free trust-funnel (equalities). [Sampaio and Toint \(2015\)](#) is the primary precedent: a derivative-free trust-funnel method that builds fully-linear models and proves first-order convergence — but for *equality* constraints $c(x) = 0$. The companion numerical study ([Sampaio and Toint, 2016](#)) reports experience extending the approach to problems “with both equality and inequality constraints and . . . simple bounds,” handling inequalities by the usual device of adding slack variables and using “neither filter nor penalty function.” Our project differs in two ways that matter for the theory: (i) we treat inequalities *directly* through the violation measure $\theta = \|[c]_+\|$ rather than by slack-variable equality reformulation, so the model geometry lives in the original x -space that the simulation samples; and (ii) our target is a *proof* of KKT convergence under MFCQ for that direct inequality treatment, not a numerical extension.

Model-based DFO with feasible inequality iterates (the contrast). NOWPAC ([Augustin and Marzouk, 2017](#)) and its stochastic generalization (S)NOWPAC ([Menhorn et al., 2022](#)) build fully-linear models of the objective and of *inequality* constraints inside a trust region — superficially the closest match. The decisive difference is that NOWPAC enforces *feasible iterates*: it adds a convex offset h_k to the constraints, the “inner boundary path,” which keeps iterates strictly inside the feasible region and is precisely what their convergence argument relies on ([Menhorn et al., 2022](#), §4). That design targets *unrelaxable* constraints (points outside $\mathcal{F}$ may not be usable) — the opposite of our Assumption 1. Requiring feasibility throughout forbids the infeasible model-sample points and normal-step excursions that our funnel both permits and needs. So NOWPAC solves an adjacent but distinct problem, and its proof technique does not transfer to the relaxable setting.

Feasibility-restoration alternatives. Inexact-restoration DFO ([Birgin et al., 2021](#)) handles constraints by alternating a feasibility-restoration phase with an optimization phase. It is a legitimate alternative architecture for relaxable constraints and a useful contrast, but it is not a trust-funnel method and does not build a single fully-linear model set used jointly by a normal and a tangential step; we mention it as related rather than as the framework we extend.

Constraint taxonomy and the DFO backdrop. [Le Digabel and Wild \(2015\)](#) give the taxonomy (quantifiable/relaxable/simulation) that names our constraint class and explains why relaxability is the enabling property. [Larson et al. \(2019\)](#) survey constrained DFO and the fully-linear-model / trust-region machinery, and [Conn et al. \(2009, Ch. 6,10,11\)](#) provide the Λ -poisedness geometry (uniform fully-linearity constants) and the unconstrained trust-region convergence template that our criticality step and $\kappa_{eg}\Delta_k \rightarrow 0$ argument specialize.

6.1 Novelty statement

To our knowledge, no existing method provides a *model-based* trust-funnel algorithm — with fully-linear models of *both* the objective and the *inequality* constraints — for *relaxable* simulation constraints, together with a proof of convergence to a KKT stationary point under MFCQ, and *without* a penalty parameter or a feasible-iterate requirement.

Concretely, the contribution combines

1. the DFO trust-funnel of [Sampaio and Toint \(2015\)](#) — but extended from *equalities* to *inequalities* via the violation measure $\theta = \|[c]_+\|$ handled *directly* in x -space;
2. the fully-linear objective-and-inequality-constraint modeling of NOWPAC ([Augustin and Marzouk, 2017](#); [Menhorn et al., 2022](#)) — but *without* its feasible-iterate inner-boundary requirement, using relaxability (Assumption 1) to sample and step through infeasible points; and
3. a KKT-under-MFCQ convergence guarantee, obtained without any penalty parameter (contrast the penalty h -functions in the repository, e.g. `general_h_funs.py`, which this project deliberately does *not* use).

6.2 Summary table

Method	DF?	FL model of c	Constraint type	Iterates	Convergence
Gould–Toint (2010)	no	—	equality	funnel	1st-order
Sampaio–Toint (2015)	yes	yes	equality	funnel	1st-order
NOWPAC / (S)NOWPAC	yes	yes	inequality	feasible-only	yes
Inexact restoration (2021)	yes	(rest.)	general	restoration	yes
This project	yes	yes	relaxable ineq.	funnel (infeasible OK)	KKT/MFCQ (target)

“FL model of c ” = fully-linear model of the constraints; “DF” = derivative-free. The last row is the deliverable of milestones M2–M3.

7 Algorithm: a DFO trust-funnel for relaxable inequalities

This section specifies the method whose convergence is analyzed in milestone M3. It keeps the five-step structure of the CSV modified-DFO algorithm as implemented in `DFO_quad_TR/Algorithm_11_1.m` (criticality $\rightarrow$ step calculation $\rightarrow$ acceptance $\rightarrow$ model improvement $\rightarrow$ radius update), and grafts onto it the normal/tangential step and funnel machinery of [Gould and Toint \(2010\)](#) and its derivative-free descendant [Sampaio and Toint \(2015\)](#), specialized to the smooth inequality infeasibility measure $\theta = \frac{1}{2}\|[c]_+\|^2$ of Section 3. The design deliberately avoids both a penalty parameter and a feasible-iterate requirement (Section 6).

7.1 Per-iteration data and constants

Entering iteration k we hold the incumbent x_k , radius Δ_k , funnel bound $\theta_k^{\max}$, and the fully-linear models $m_f^k, m_{c,i}^k$ of Section 2 with gradient $g_k = \nabla m_f^k(x_k)$ and Jacobian $J_k = [\nabla m_{c,i}^k(x_k)]$. Write $c_k = c(x_k)$, $f_k = f(x_k)$, $\theta_k = \theta(x_k) = \frac{1}{2}\|[c_k]_+\|^2$. Because f and c come from one vector blackbox $F = (f, c_1, \dots, c_p)$, all $p + 1$ models are produced by a single call to `formquad` sharing the same Λ -poised sample set, and the `valid` flag is the computational witness of Assumption 5.

The constants below reuse the names in `DF0options.mat` / `Algorithm_11_1.m`; the funnel/step constants are new and named in the Gould–Toint style. All are fixed over the run.

$\eta_0 \in (0, \eta_1)$,	$\eta_1 \in (0, 1)$	(ratio-test thresholds)	<code>eta_0, eta_1</code>
$\gamma \in (0, 1)$,	$\gamma_{\text{inc}} > 1$	(radius shrink / expand)	<code>gamma, gamma_inc</code>
$\Delta_{\text{max}} > 0$,	$\Delta_{\text{min}} > 0$	(radius cap / stop)	<code>Delta_max, Delta_min</code>
$\epsilon_c > 0$,	$\omega \in (0, 1)$	(criticality threshold / shrink)	<code>epsilon_c, omega</code>
$\mu > 0$,	$\beta > 0$	(criticality–radius coupling)	<code>mu, beta</code>
$r \geq 1$		(poisedness-radius factor)	<code>r</code>
$\kappa_n \in (0, 1]$		(normal-step radius fraction)	<code>new</code>
ω_n, ω_t		(normal-trigger / switching bounding functions)	<code>new</code>
$\kappa_\omega \in (0, 1)$		(trigger/switching consistency)	<code>new</code>
$\kappa_{\text{tg}} \in (0, 1)$		(tangential-objective domination)	<code>new</code>
$\kappa_{\text{nt}} \in (0, 1)$		(feasibility-retention relaxation)	<code>new</code>
$\kappa_{\text{ca}} \in (0, 1), \kappa_{\text{cb}} \in (0, 1)$		(funnel-update constants)	<code>new</code>
$\kappa_\varepsilon > 0$		(active-set tolerance scale)	<code>new</code>

Here $\omega_n, \omega_t : [0, \infty) \rightarrow [0, \infty)$ are continuous, non-decreasing *bounding functions* with $\omega_n(t) = 0 \iff t = 0$ and ω_t strictly increasing near 0; the canonical choice is $\omega_n(t) = \omega_t(t) = \kappa t^q$ for constants $\kappa > 0, q \geq 1$.

Standing constant relations (used in M3). The analysis requires the following relations, all satisfiable by construction:

1. *Feasibility radius cap*: on normal-step iterations the model management enforces $\Delta_k \leq \mu \chi_k^c$ (14), so (with L1 and cq-nonstat) $\chi_k^c \geq \kappa_m \sqrt{2\theta_k} / (1 + \mu \kappa_{\text{eg}}^\theta)$: an over-large radius cannot make the model report a spuriously small feasibility criticality. (With the smooth measure this replaces the ϵ_c -vs- κ_m relation of the unsquared draft.)
2. *Trigger/switching consistency* (Gould and Toint, 2010, (2.31)): $\omega_t(\omega_n(t)) \leq \kappa_\omega t$ for all $t \geq 0$. This couples the normal-step trigger (15) and the switching test (19) so that they cannot both be dormant while $s_k \neq 0$ is productive (the residual $s_k = 0$ case is the μ -iteration of Step 3, controlled by Lemma 10).
3. *Shrinking active-set tolerance*: the model active set uses $\varepsilon_k = \kappa_\varepsilon \sigma_k \rightarrow 0$ (rather than a fixed ε), so $\mathcal{A}_k(\varepsilon_k)$ identifies the true active set at a limit point (needed for limiting complementarity in T2).

7.2 Step 1 — Criticality step

Define the aggregate criticality

$$\sigma_k = \max(\chi_k^c, \pi_k^f), \quad (12)$$

with χ_k^c, π_k^f from (8)–(9). This is the constrained analog of the scalar `sigma` = $\max(\|c_{\text{icb}}\|, -\lambda_{\min}(Q))$ used in `Algorithm_11_1.m`: it collapses “how nonstationary are we” for both feasibility and optimality into one number.

If $\sigma_k > \epsilon_c$, skip to Step 2 with the incumbent models and Δ_k . Otherwise invoke the criticality loop (the analog of `Algorithm_10_2.m`): while the models are not certifiably fully linear on

$B(x_k, r\Delta_k)$ or $r\Delta_k > \mu\sigma_k$, apply a Λ -poised model-improvement step on $B(x_k, \omega^i\Delta_k)$, rebuild $m_f^k, m_{c,i}^k$, recompute σ_k , and set $\Delta_k \leftarrow \omega^i\Delta_k$. On exit set

$$\Delta_k \leftarrow \min(\max(\Delta_k, \beta\sigma_k), \Delta_k^{\text{in}}), \quad (13)$$

matching the final line of **Algorithm_10_2.m**. If $\Delta_k < \Delta_{\min}$, stop (approximate stationarity). This is the step that *ties* Δ_k to σ_k and is what lets model stationarity transfer to true stationarity in M3; it is the ingredient that “fully-linear models on every iteration” does not supply on its own.

Feasibility radius cap on normal-step iterations. Whenever a normal step will be computed (Step 2a), the model-management additionally enforces

$$\Delta_k \leq \mu\chi_k^c, \quad (14)$$

i.e. the trust region is sized to the *feasibility* criticality it is about to reduce (shrinking Δ_k and rebuilding as in the criticality loop if (14) is violated). This is the feasibility analog of the standard criticality step (Conn et al., 2009, Ch. 10): it prevents an over-large radius from letting the fully-linear model report a spuriously small χ_k^c while the true infeasibility criticality is large. With (14) and L1, cq-nonstat yields $\chi_k^c \geq \kappa_m \sqrt{2\theta_k}/(1 + \mu\kappa_{\text{eg}}^\theta)$ on every normal-step iteration (used throughout the feasibility analysis, M3).

7.3 Step 2 — Step calculation (normal then tangential)

The trial step is a composite $s_k = n_k + t_k$ kept in $B(x_k, \Delta_k)$.

(2a) Normal step n_k — reduce linearized infeasibility. A normal step is computed when the incumbent infeasibility is significant *relative to the optimality measure of the previous iterate*, or when the current point is too infeasible to pursue optimality (switching (19) fails), i.e. when

$$\theta_k > 0 \quad \text{and} \quad \left(\theta_k \geq \omega_n(\pi_{k-1}^f) \quad \text{or} \quad \pi_k^f \leq \omega_t(\theta_k) \right), \quad (15)$$

and $n_k = 0$ otherwise. The second disjunct (switching failure) guarantees that *every* c -iteration carries a genuine normal step, which the feasibility analysis (L8) needs; the first is the optimality-forcing trigger of Gould and Toint (2010, (2.30)), **the correct replacement for a radius-based one**. A trigger such as $\theta_k > \kappa\Delta_k^2$ is a rate error (near a critical point both $\theta_k \rightarrow 0$ and $\Delta_k \rightarrow 0$, and $\theta_k > \kappa\Delta_k^2$ can persist, e.g. $\theta_k \asymp \Delta_k \gg \Delta_k^2$, never switching off); a purely *funnel*-relative trigger $\theta_k > \kappa\theta_k^{\max}$ can instead *stall* at an infeasible, non- θ -stationary point where the funnel is frozen. Tying the trigger to π_{k-1}^f via ω_n (with $\omega_n(t) = 0 \iff t = 0$ and the consistency $\omega_t(\omega_n(t)) \leq \kappa_\omega t$) forces feasibility *as optimality improves*: as $\pi_{k-1}^f \rightarrow 0$, even a tiny θ_k triggers a normal step. This coupling is what drives $\theta_k \rightarrow 0$ in T1. When triggered, n_k approximately solves the convex (piecewise-linear/quadratic) trust-region subproblem in the (smooth) linearized violation,

$$\min_{n \in \mathbb{R}^n} \frac{1}{2} \| [c_k + J_k n]_+ \|^2 \quad \text{s.t.} \quad \|n\| \leq \kappa_n \Delta_k, \quad (16)$$

and is required to achieve a Cauchy-type decrease of the model infeasibility: there is $\kappa_{\text{nc}} > 0$ with

$$\theta_k - \theta_k^m(x_k + n_k) \geq \kappa_{\text{nc}} \chi_k^c \min\left(\frac{\chi_k^c}{1 + \beta_\theta}, \kappa_n \Delta_k\right), \quad (17)$$

where β_θ bounds the curvature of θ_k^m (finite, since $\|J_k\| \leq \kappa_J$). This is the standard fraction-of-Cauchy decrease for the smooth θ_k^m along the steepest-descent direction $-\nabla\theta_k^m(x_k)$ (the objective

of (16) is convex, so a generalized Cauchy point attains it). The normal step is also required to be *normal*,

$$\|n_k\| \leq \kappa_{nn} \|[c_k]_+\| = \kappa_{nn} \sqrt{2\theta_k} \quad (\kappa_{nn} > 0), \quad (18)$$

the standard trust-funnel requirement (Gould and Toint, 2010, §2) that makes n_k vanish as $\theta_k \rightarrow 0$ (used in Lemma 15). Computationally (16) is a small bound-constrained convex QP in (c_k, J_k) — the role played by `minqsw/bqmin` in the code-bridge (D5).

(2b) Tangential step t_k — reduce the objective model. Given n_k , the tangential step reduces the objective model along directions that do not undo the linearized feasibility gain. Let $\mathcal{A}_k = \mathcal{A}_k(\varepsilon_k)$ be the model active set (Section 4) with the *shrinking* tolerance $\varepsilon_k = \kappa_\varepsilon \sigma_k$, and N_k a basis for (an approximation of) the null space $\{d : J_k^A d = 0\}$. We take a tangential step only when the optimality criticality is significant relative to the current infeasibility — the *switching condition* (Gould and Toint, 2010, (2.29))

$$\pi_k^f > \omega_t(\theta_k) \quad (\text{equivalently } \theta_k < \omega_t^{-1}(\pi_k^f): \text{ infeasibility small relative to } \pi_k^f). \quad (19)$$

When (19) holds, t_k approximately solves

$$\min_t m_f^k(x_k + n_k + t) \quad \text{s.t.} \quad J_k^A t \leq 0, \quad \|n_k + t\| \leq \Delta_k, \quad \theta_k^m(x_k + n_k + t) \leq \vartheta_k, \quad (20)$$

where the last constraint — the analog of Gould and Toint (2010, (2.28)) for the squared measure — caps the model infeasibility of the composite step at

$$\vartheta_k := \kappa_{nt} \theta_k + (1 - \kappa_{nt}) \theta_k^m(x_k + n_k) \quad (\kappa_{nt} \in (0, 1)), \quad (21)$$

so the tangential step cannot undo the normal step’s feasibility gain by more than a fraction. (It is feasible: $t = 0$ satisfies it since $\theta_k^m(x_k + n_k) \leq \vartheta_k$.) This *feasibility-retention* bound lets the composite predicted decrease inherit the normal-step Cauchy bound: $\text{pred}_k^\theta = \theta_k - \theta_k^m(x_k + s_k) \geq \theta_k - \vartheta_k = (1 - \kappa_{nt})(\theta_k - \theta_k^m(x_k + n_k))$, i.e. $\text{pred}_k^\theta \geq (1 - \kappa_{nt}) \times (17)$ (used in L8). On this step t_k must deliver a fraction-of-Cauchy decrease of the objective model on the tangent space: there is $\kappa_{tc} > 0$ with

$$m_f^k(x_k + n_k) - m_f^k(x_k + s_k) \geq \kappa_{tc} \pi_k^f \min\left(\frac{\pi_k^f}{1 + \|H_k\|}, \Delta_k\right), \quad (22)$$

where H_k is the objective model Hessian. If (19) fails, set $t_k = 0$ (this is a pure feasibility, or “c-”, iteration). In all cases $s_k = n_k + t_k$ with $\|s_k\| \leq \Delta_k$. When $p = 0$ and $\theta_k \equiv 0$ this reduces exactly to the unconstrained trust-region step of Algorithm_11_1.m (Step 2, `s_k = trust(c,Q,Delta)`).

7.4 Step 3 — Iteration type and acceptance

Define the predicted decreases

$$\text{pred}_k^f = m_f^k(x_k) - m_f^k(x_k + s_k), \quad (23)$$

$$\text{pred}_k^\theta = \theta_k - \theta_k^m(x_k + s_k), \quad (24)$$

and the achieved decreases $\text{ared}_k^f = f_k - f(x_k + s_k)$, $\text{ared}_k^\theta = \theta_k - \theta(x_k + s_k)$, computed from a single blackbox evaluation of F at the trial point $x_k + s_k$ (yielding f and all c_i at once). The iteration is classified into three mutually exclusive types (following Gould and Toint, 2010), the μ -iteration handling the degenerate $s_k = 0$ case that a two-way f/c split would leave undefined:

μ -iteration if $s_k = 0$. Under the strengthened trigger (15) this is a *model-critical* situation: $t_k = 0$ with switching held would force $\pi_k^f = 0$ (L3), contradicting (19); hence switching failed ($\pi_k^f \leq \omega_t(\theta_k)$), the trigger's second disjunct fired, and $n_k = 0$ then forces $\chi_k^c = 0$ (the normal Cauchy bound (17) is null). So a μ -iteration has $\chi_k^c = 0$ and $\pi_k^f \leq \omega_t(\theta_k)$ (Lemma 10); by cq-nonstat (39), $\chi_k^c = 0$ forces $\sqrt{2\theta_k} \leq \kappa_{\text{eg}}^\theta \Delta_k / \kappa_m$, i.e. the point is *near-feasible relative to the radius*. No trial point is evaluated; the iteration is routed through the criticality step (Step 1), the funnel is unchanged, and $x_{k+1} = x_k$. In particular, a μ -iteration *cannot occur at a point with $\theta_k \geq \tau > 0$ and $\Delta_k < \kappa_m \sqrt{2\tau} / \kappa_{\text{eg}}^\theta$* — the fact the feasibility analysis (L8) relies on.

f -iteration if $s_k \neq 0$, a tangential step was computed ((19) held), and the full objective decrease retains a fraction of the *tangential* objective decrease,

$$\text{pred}_k^f \geq \kappa_{\text{tg}} \delta_k^{f,t}, \quad \delta_k^{f,t} := m_f^k(x_k + n_k) - m_f^k(x_k + s_k) \quad (25)$$

(the Gould and Toint (2010, (2.33)) domination test; $\delta_k^{f,t}$ is the quantity bounded below by L3, so (25) transfers that bound to pred_k^f — this is what the feasibility path-length lemma L6 uses, and it replaces the earlier, incorrect comparison of pred_k^f to pred_k^θ). Acceptance uses the objective ratio

$$\rho_k^f = \frac{\text{ared}_k^f}{\text{pred}_k^f}. \quad (26)$$

The trial point is accepted iff it respects the funnel and the ratio passes:

$$\theta(x_k + s_k) \leq \theta_k^{\max} \quad \text{and} \quad (\rho_k^f \geq \eta_1, \text{ or } (\rho_k^f > \eta_0 \text{ and models valid})). \quad (27)$$

The funnel bound is left unchanged: $\theta_{k+1}^{\max} = \theta_k^{\max}$.

c -iteration otherwise ($s_k \neq 0$ and (25) fails or no tangential step was computed). A *c-iteration always carries a genuine normal step* ($n_k \neq 0$ whenever $\chi_k^c > 0$): if switching (19) failed, the trigger (15) computed n_k directly; if switching held but (25) failed, then $\text{pred}_k^f < \kappa_{\text{tg}} \delta_k^{f,t} \leq \delta_k^{f,t}$, which (as $\text{pred}_k^f = \delta_k^{f,t} + [m_f^k(x_k) - m_f^k(x_k + n_k)]$) forces $m_f^k(x_k) - m_f^k(x_k + n_k) < 0$, i.e. $n_k \neq 0$. This closes the gap that a pure tangential step could be classified *c* without a normal step. Acceptance uses the feasibility ratio

$$\rho_k^c = \frac{\text{ared}_k^\theta}{\text{pred}_k^\theta}, \quad (28)$$

well defined since the normal step gives, via retention (21), $\text{pred}_k^\theta \geq (1 - \kappa_{\text{nt}}) \times (17) > 0$. The trial point is accepted iff $\rho_k^c \geq \eta_1$. On a successful *c*-iteration the funnel is tightened (Step 4b).

On acceptance, $x_{k+1} = x_k + s_k$; on rejection, $x_{k+1} = x_k$. As in `Algorithm_11_1.m`, the trial point (and its already-paid blackbox value) is *always* added to the sample set regardless of acceptance, since evaluations are expensive.

7.5 Step 4 — Model improvement and funnel update

(4a) Model improvement. If the iteration was unsuccessful ($\rho_k^\bullet < \eta_1$) or the step was negligible ($\|s_k\|/\Delta_k$ below `tol_step_ratio`), run a Λ -poised model-improvement step on $B(x_k, r\Delta_k)$ and rebuild the models — exactly Step 4 of `Algorithm_11_1.m` via `Model_Improvement_Step.m`.

(4b) Funnel update. On a successful c -iteration, tighten the funnel by the Gould–Toint rule

$$\theta_{k+1}^{\max} = \max\left(\kappa_{\text{ca}} \theta_k^{\max}, \theta_{k+1} + \kappa_{\text{cb}} (\theta_k^{\max} - \theta_{k+1})\right), \quad (29)$$

which keeps $\theta_{k+1} < \theta_{k+1}^{\max} \leq \theta_k^{\max}$ (strict room below the cap, monotone non-increasing cap). On f -iterations and unsuccessful iterations, $\theta_{k+1}^{\max} = \theta_k^{\max}$. This monotonicity is lemma L4 of M3 and is the mechanism that forces $\theta_k \rightarrow 0$.

7.6 Step 5 — Trust-region radius update

Mirroring Step 5 of `Algorithm_11_1.m`:

$$\Delta_{k+1} = \begin{cases} \min(\gamma_{\text{inc}} \Delta_k, \Delta_{\max}) & \text{very successful } (\rho_k^\bullet \geq \eta_1) \text{ and } \|s_k\| \approx \Delta_k, \\ \gamma \Delta_k & \text{unsuccessful } (\rho_k^\bullet < \eta_1) \text{ and models valid on } B(x_k, r\Delta_k), \\ \Delta_k & \text{otherwise,} \end{cases} \quad (30)$$

where $\rho_k^\bullet$ is ρ_k^f on f -iterations and ρ_k^c on c -iterations. Increment k and return to Step 1.

7.7 Consolidated pseudocode

Algorithm 1 (DFO trust-funnel for relaxable inequalities).

Given feasible x_0 (Assumption 4), Δ_0 , $\theta_0^{\max} \geq \theta_0$, constants of Section 7.1.

For $k = 0, 1, 2, \dots$:

1. **Models.** Build fully-linear $m_f^k, \{m_{c,i}^k\}$ on $B(x_k, \Delta_k)$; form g_k, J_k ; compute $\chi_k^c, \pi_k^f, \sigma_k = \max(\chi_k^c, \pi_k^f)$.
2. **Criticality.** If $\sigma_k \leq \epsilon_c$: run the `Algorithm_10_2` loop (improve poisedness, shrink Δ_k toward $\mu\sigma_k$, apply (13)); if $\Delta_k < \Delta_{\min}$ **stop**.
3. **Normal step.** If $\theta_k \geq \omega_n(\pi_{k-1}^f)$, compute n_k solving (16) with decrease (17); else $n_k = 0$.
4. **Tangential step.** If switching (19) holds, compute t_k solving (20) with decrease (22); else $t_k = 0$. Set $s_k = n_k + t_k$ ($\|s_k\| \leq \Delta_k$).
5. **Classify & evaluate.** If $s_k = 0$: μ -iteration — criticality/geometry refresh, $x_{k+1} = x_k$, funnel unchanged, go to Step 7. Else evaluate $F(x_k + s_k)$ and classify f - vs. c -iteration (Step 3); accept/reject by (27)/(28).
6. **Improve & funnel.** Model improvement (4a) if needed; funnel update (29) on a successful c -iteration.
7. **Radius.** Update Δ_{k+1} by (30).

Design invariants (to be discharged in M3). (i) $\theta_k \leq \theta_k^{\max}$ for all k and $\{\theta_k^{\max}\}$ is non-increasing; (ii) every accepted step lies in $B(x_k, \Delta_k)$; (iii) with $p = 0$ (so $\theta_k \equiv 0$, $n_k \equiv 0$, always f -iterations) Algorithm 1 collapses to `Algorithm_11_1.m`; (iv) rewriting an equality $c^{\text{eq}}(x) = 0$ as $\{c^{\text{eq}} \leq 0, -c^{\text{eq}} \leq 0\}$ recovers the equality trust-funnel of Sampaio and Toint (2015). These invariants are the sanity checks the convergence analysis (M3) must respect.

8 Convergence analysis

Throughout, Assumptions 1–6 hold. We write $\{x_k\}$ for the incumbents produced by Algorithm 1, $\S$ for the index set of *successful* iterations (those on which $x_{k+1} = x_k + s_k$), and we say a model

quantity is “valid at k ” when the models are fully linear on $B(x_k, \Delta_k)$ (Assumption 5). Two consequences of the standing assumptions are used repeatedly and recorded first.

Proposition 1 (Boundedness). *f is bounded below on the working region $\mathcal{L}$ (Assumption 3), and there is $\kappa_H \geq 0$ with $\|H_k\| \leq \kappa_H$ and $\|J_k\| \leq \kappa_J$ for all k (model Hessian of f and model Jacobian of c are uniformly bounded).*

Proof. $\mathcal{L}$ is bounded (Assumption 3) and $f, c \in C^1$ (Assumption 2), so f attains a finite minimum on $\bar{\mathcal{L}}$. Fully-linear gradients satisfy $\|g_k\| \leq \|\nabla f(x_k)\| + \kappa_{\text{eg}} \Delta_k$ and likewise for J_k ; since $\nabla f, \nabla c$ are bounded on $\mathcal{L}$ and $\Delta_k \leq \Delta_{\max}$, both are uniformly bounded. Minimum-Frobenius-norm model Hessians (as returned by `formquad`) are bounded by the same data. $\square$

Standing constant relations. The analysis uses the relations fixed in Section 7.1: the active-set tolerance shrinks, $\varepsilon_k = \kappa_\varepsilon \sigma_k \rightarrow 0$ (for limiting complementarity in T2); and, for the smooth measure, Lemma 11 supplies a feasibility-criticality lower bound $\chi_k^c \geq \kappa_m \sqrt{2\theta_k} - \kappa_{\text{eg}}^\theta \Delta_k$ whose “in particular” clause is invoked with $\varrho = \tau/2$ (a fixed positive constant in the T1 contradiction). No lower bound on ϵ_c relative to the (unknown) infeasibility level is needed: the radius controls come from the reproduced lemmas of Section 8.2 and the criticality step (L5).

8.1 Supporting lemmas

L1 — Model-error propagation (objective, constraints, and θ).

Lemma 1 (L1). *Let the models be valid at k . Then for all $y \in B(x_k, \Delta_k)$,*

$$|f(y) - m_f^k(y)| \leq \kappa_{\text{ef}} \Delta_k^2, \quad \|\nabla f(y) - \nabla m_f^k(y)\| \leq \kappa_{\text{eg}} \Delta_k, \quad (31)$$

$$|c_i(y) - m_{c,i}^k(y)| \leq \kappa_{\text{ef}} \Delta_k^2, \quad \|\nabla c_i(y) - \nabla m_{c,i}^k(y)\| \leq \kappa_{\text{eg}} \Delta_k, \quad (32)$$

and, with the smooth $\theta = \frac{1}{2} \| [c]_+ \|^2$ and its model $\theta_k^m = \frac{1}{2} \| [m_c^k]_+ \|^2$, θ has a fully-linear model on $B(x_k, \Delta_k)$:

$$|\theta(y) - \theta_k^m(y)| \leq \kappa_{\text{ef}}^\theta \Delta_k^2, \quad \|\nabla \theta(y) - \nabla \theta_k^m(y)\| \leq \kappa_{\text{eg}}^\theta \Delta_k, \quad (33)$$

with $\kappa_{\text{ef}}^\theta = \sqrt{p} C_\theta \kappa_{\text{ef}}$ and $\kappa_{\text{eg}}^\theta = \sqrt{p} (C_\theta \kappa_{\text{eg}} + \kappa_J \kappa_{\text{ef}} \Delta_{\max})$, where $C_\theta := \sup_{\mathcal{L}} \| [c]_+ \| + \sqrt{p} \kappa_{\text{ef}} \Delta_{\max}^2 < \infty$.

Proof. (31)–(32) are Definition 1 applied to f and each c_i . For (33), write $u = [c(y)]_+$, $v = [m_c^k(y)]_+$; the map $z \mapsto [z]_+$ is 1-Lipschitz, so $\|u - v\| \leq \|c(y) - m_c^k(y)\| \leq \sqrt{p} \kappa_{\text{ef}} \Delta_k^2$ by (32). Then $|\theta - \theta_k^m| = \frac{1}{2} |\|u\|^2 - \|v\|^2| = \frac{1}{2} |\langle u + v, u - v \rangle| \leq \frac{1}{2} (\|u\| + \|v\|) \|u - v\| \leq C_\theta \|u - v\|$, giving the value bound. For the gradient, $\nabla \theta(y) = J(y)^\top [c(y)]_+$ and $\nabla \theta_k^m(y) = J_k^\top [m_c^k(y)]_+$; adding and subtracting $J_k^\top [c(y)]_+$ and using (32), $\|[c(y)]_+\| \leq C_\theta$, and $\|J_k\| \leq \kappa_J$ gives the $O(\Delta_k)$ bound. All constants are k -independent (Assumption 5, Proposition 1). $\square$

The point of L1 is that *the smooth infeasibility measure inherits a fully-linear model* — $O(\Delta_k^2)$ in value and $O(\Delta_k)$ in gradient — from the constraint models. This lets the feasibility analysis treat θ as a smooth merit with a fully-linear model, so the reproduced trust-region lemmas of Section 8.2 apply to it verbatim; it is why fully-linear models of the *constraints* (not merely the objective) are required.

L2 — Normal-step decrease (feasibility).

Lemma 2 (L2). *Suppose a normal step is computed at k and n_k is a generalized Cauchy point of the convex subproblem (16). Then, with β_θ a curvature bound for θ_k^m ,*

$$\theta_k - \theta_k^m(x_k + n_k) \geq \kappa_{nc} \chi_k^c \min\left(\frac{\chi_k^c}{1+\beta_\theta}, \kappa_n \Delta_k\right), \quad \kappa_{nc} = \frac{1}{2}. \quad (34)$$

Proof. This is the classical fraction-of-Cauchy decrease for the *smooth* convex θ_k^m minimized along its steepest-descent direction $-\nabla\theta_k^m(x_k)$ inside $\|n\| \leq \kappa_n \Delta_k$: with criticality $\chi_k^c = \|\nabla\theta_k^m(x_k)\|$ and curvature $\leq \beta_\theta$, the Cauchy point gives $\theta_k^m(x_k) - \theta_k^m(x_k + n_k^C) \geq \frac{1}{2}\chi_k^c \min(\chi_k^c/\beta_\theta, \kappa_n \Delta_k)$ (Conn et al., 2000, Ch. 6.3; the same bound underlies (TR-acc) of Section 8.2). A generalized Cauchy point does at least this well. β_θ is finite: $\theta_k^m = \frac{1}{2}\|m_c^k\|_+^2$ has Hessian (where differentiable) $J_k^\top D J_k + \sum_i [m_{c,i}^k]_+ \nabla^2 m_{c,i}^k$ bounded by Proposition 1. $\square$

Role of MFCQ. (34) is vacuous unless $\chi_k^c > 0$, i.e. unless θ_k^m has nonzero gradient. MFCQ (Assumption 6) is exactly what guarantees $\|\nabla\theta\| > 0$ at infeasible points; this is made precise for the smooth measure in Lemma 11 and is the pivot of the feasibility theorem T1.

L3 — Tangential-step decrease (optimality).

Lemma 3 (L3). *Suppose the switching condition (19) holds at k and t_k is a (projected) generalized Cauchy point of (20). Then*

$$m_f^k(x_k + n_k) - m_f^k(x_k + s_k) \geq \kappa_{tc} \pi_k^f \min\left(\frac{\pi_k^f}{1+\kappa_H}, \Delta_k\right), \quad \kappa_{tc} = \frac{1}{2}. \quad (35)$$

Proof. This is the fraction-of-Cauchy-decrease bound for a quadratic model minimized over the polyhedral cone $\{t : J_k^A t \leq 0\}$ intersected with the trust region, with π_k^f the projected-gradient criticality (9). The argument is the constrained Cauchy-point estimate of Conn et al. (2009, Ch. 10) (their unconstrained $\|g_k\|$ replaced by the projected measure π_k^f and the line search restricted to the feasible cone), using $\|H_k\| \leq \kappa_H$ from Proposition 1. When $\mathcal{A}_k = \emptyset$ the cone is all of $\mathbb{R}^n$, $\pi_k^f = \|g_k\|$, and (35) is precisely the unconstrained Cauchy decrease of Algorithm_11_1.m. $\square$

L4 — Funnel monotonicity.

Lemma 4 (L4). *The funnel sequence $\{\theta_k^{\max}\}$ is non-increasing and bounded below by 0, hence convergent to some $\theta_*^{\max} \geq 0$. Every iterate satisfies $\theta_k \leq \theta_k^{\max}$, and on every successful c -iteration*

$$\theta_k^{\max} - \theta_{k+1}^{\max} \geq (1 - \kappa_{cb}) (\theta_k^{\max} - \theta_{k+1}). \quad (36)$$

Proof. By construction (29) fires only on successful c -iterations and returns $\theta_{k+1}^{\max} = \max(\kappa_{ca} \theta_k^{\max}, \theta_{k+1} + \kappa_{cb}(\theta_k^{\max} - \theta_{k+1}))$; on all other iterations $\theta_{k+1}^{\max} = \theta_k^{\max}$. Since a successful c -iteration has $\rho_k^c \geq \eta_1 > 0$, hence $\text{ared}_k^\theta > 0$ and $\theta_{k+1} < \theta_k \leq \theta_k^{\max}$, both branches of the max are $\leq \theta_k^{\max}$ (as $\kappa_{ca}, \kappa_{cb} \in (0, 1)$), so the sequence is non-increasing; it is bounded below by 0, so it converges. Acceptance rules (27)/(28) never admit a point with $\theta > \theta_k^{\max}$ (the f -branch enforces it explicitly; the c -branch decreases θ from $\theta_k \leq \theta_k^{\max}$), giving $\theta_k \leq \theta_k^{\max}$ for all k . Finally, from the second argument of the max, $\theta_{k+1}^{\max} \leq \theta_{k+1} + \kappa_{cb}(\theta_k^{\max} - \theta_{k+1})$, which rearranges to (36). $\square$

8.2 Reproduced single-region trust-region lemmas

We reproduce, self-contained, the standard single-region trust-region results (Conn et al., 2000, Ch. 6.4; Conn et al., 2009, Ch. 10) in the exact abstract form the constrained analysis invokes, so nothing here is imported. The *abstract setting* is one iteration acting on a merit φ (later f or θ) via a model m fully linear on $B(x_k, \Delta_k)$ with value error $\leq \kappa_{\text{ef}}^\varphi \Delta_k^2$ (Definition 1); a step $\|s_k\| \leq \Delta_k$ with model decrease obeying a *fraction-of-Cauchy* bound

$$\text{pred}_k := m(x_k) - m(x_k + s_k) \geq \kappa_{\text{fcd}} \xi_k \min\left(\frac{\xi_k}{1+\kappa_H^\varphi}, \Delta_k\right) \quad (37)$$

for a criticality measure $\xi_k \geq 0$ and curvature bound κ_H^φ ; ratio $\rho_k = \text{ared}_k / \text{pred}_k$ with $\text{ared}_k = \varphi(x_k) - \varphi(x_k + s_k)$; acceptance iff $\rho_k \geq \eta_1$; and the radius update (30) (shrink by γ only on unsuccessful iterations; never shrink on a successful one). All of L2, L3 supply (37) (for θ with $\xi = \chi^c$, and for f with $\xi = \pi^f$ respectively).

Lemma 5 (TR-acc — accuracy implies success). *In the abstract setting, if $\Delta_k \leq \min(\frac{\xi_k}{1+\kappa_H^\varphi}, \frac{\kappa_{\text{fcd}}(1-\eta_1)\xi_k}{2\kappa_{\text{ef}}^\varphi})$ then $\rho_k \geq \eta_1$ (the iteration is successful).*

Proof. Since $x_k, x_k + s_k \in B(x_k, \Delta_k)$, full-linearity gives $|\text{ared}_k - \text{pred}_k| \leq 2\kappa_{\text{ef}}^\varphi \Delta_k^2$. The first bound on Δ_k makes the min in (37) equal Δ_k , so $\text{pred}_k \geq \kappa_{\text{fcd}} \xi_k \Delta_k > 0$. Hence $|\rho_k - 1| \leq 2\kappa_{\text{ef}}^\varphi \Delta_k^2 / (\kappa_{\text{fcd}} \xi_k \Delta_k) = 2\kappa_{\text{ef}}^\varphi \Delta_k / (\kappa_{\text{fcd}} \xi_k) \leq 1 - \eta_1$ by the second bound, so $\rho_k \geq \eta_1$. $\square$

Lemma 6 (TR1 — radius lower bound at non-critical points). *If $\xi_j \geq \varsigma > 0$ for every iteration j in an index interval $[k_0, k_1]$ on which the model is fully linear, then $\Delta_j \geq \Delta_{\text{lb}}(\varsigma) := \gamma \min(\frac{\varsigma}{1+\kappa_H^\varphi}, \frac{\kappa_{\text{fcd}}(1-\eta_1)\varsigma}{2\kappa_{\text{ef}}^\varphi}, \Delta_{k_0})$ for all $j \in [k_0, k_1]$.*

Proof. Let $\bar{\Delta}(\varsigma) = \min(\frac{\varsigma}{1+\kappa_H^\varphi}, \frac{\kappa_{\text{fcd}}(1-\eta_1)\varsigma}{2\kappa_{\text{ef}}^\varphi})$. If $\Delta_j \leq \bar{\Delta}(\varsigma) \leq \bar{\Delta}(\xi_j)$ then TR-acc makes iteration j successful, so $\Delta_{j+1} \geq \Delta_j$ ((30) never shrinks on success). Thus the radius can drop below $\bar{\Delta}(\varsigma)$ only by a single γ -shrink from a value $> \bar{\Delta}(\varsigma)$, landing $\geq \gamma \bar{\Delta}(\varsigma)$; and it starts at Δ_{k_0} . Hence $\Delta_j \geq \gamma \min(\bar{\Delta}(\varsigma), \Delta_{k_0})$ throughout. $\square$

Lemma 7 (TR2 — liminf criticality). *Suppose φ is bounded below on $\mathcal{L}$ and is non-increasing over the iterations under consideration (each successful one decreasing it by $\geq \eta_1 \text{pred}$, the others leaving x fixed). Then $\liminf_k \xi_k = 0$.*

Proof. Suppose $\xi_k \geq \varsigma > 0$ for all large k . If there are infinitely many successful iterations, each decreases φ by $\geq \eta_1 \kappa_{\text{fcd}} \varsigma \min(\varsigma / (1 + \kappa_H^\varphi), \Delta_k) \geq \eta_1 \kappa_{\text{fcd}} \varsigma \min(\varsigma / (1 + \kappa_H^\varphi), \Delta_{\text{lb}}(\varsigma)) > 0$ (TR1), so $\varphi \rightarrow -\infty$, contradicting boundedness. If instead there are only finitely many successful iterations, then past the last one every iteration is unsuccessful, so (30) shrinks $\Delta_k \rightarrow 0$; but once $\Delta_k \leq \bar{\Delta}(\varsigma)$, TR-acc forces success — a contradiction. Either way $\xi_k \geq \varsigma$ eventually is impossible, so $\liminf_k \xi_k = 0$. $\square$

Lemma 8 (TR-lim — full criticality limit under the criticality step). *If, in addition, the criticality step ties Δ_k to ξ_k (Lemma 9: $\Delta_k \geq \kappa_\Delta \xi_k$ near non-critical points and $\Delta_k \rightarrow 0$ only as $\xi_k \rightarrow 0$), then $\lim_{k \rightarrow \infty} \xi_k = 0$.*

Proof. Standard “eventually-successful” argument of Conn et al. (2000, Thm 6.4.6) / Conn et al. (2009, Ch. 10): were $\xi_k \geq 2\varsigma$ along a subsequence while (by TR2) $\xi_k < \varsigma$ infinitely often, the criticality change between a $\xi < \varsigma$ index and the next $\xi \geq 2\varsigma$ index is bounded below, forcing a total φ -decrease that (via TR1 and φ bounded below) can occur only finitely often — a contradiction. Hence $\lim \xi_k = 0$. $\square$

L5 — Finiteness of the criticality step.

Lemma 9 (L5). *If $\sigma_k > 0$, the criticality loop of Step 2 (Section 7.2) terminates after finitely many model-improvement steps, and on exit*

$$\Delta_k \geq \min(\Delta_k^{\text{in}}, \beta\sigma_k) \quad \text{and} \quad \Delta_k \leq \mu\sigma_k \quad \text{or} \quad \Delta_k = \Delta_k^{\text{in}}. \quad (38)$$

Consequently there is $\kappa_\Delta > 0$ with $\Delta_k \geq \kappa_\Delta \sigma_k$ whenever the criticality step is entered.

Proof. For a Λ -poised set, finitely many model-improvement steps produce a fully-linear model on any given radius (Conn et al., 2009, Ch. 6, Theorem on model improvement). The loop shrinks the radius geometrically by $\omega \in (0, 1)$ and exits once the model is fully linear on $B(x_k, r\Delta_k)$ and $r\Delta_k \leq \mu\sigma_k$; since $\omega^i \Delta_k^{\text{in}} \rightarrow 0 < \mu\sigma_k/r$ (as $\sigma_k > 0$) and each radius needs only finitely many improvement steps, the loop terminates. The exit assignment (13), $\Delta_k = \min(\max(\Delta_k, \beta\sigma_k), \Delta_k^{\text{in}})$, gives the lower bound in (38); the exit test gives the upper alternative. For the final bound, either $\Delta_k = \Delta_k^{\text{in}}$ (the loop did not shrink, so $\Delta_k \geq \min(\Delta_k^{\text{in}}, \beta\sigma_k)$ trivially) or the exit assignment gave $\Delta_k \geq \beta\sigma_k$; in both cases $\Delta_k \geq \beta\sigma_k$ whenever the criticality step is entered ($\sigma_k \leq \epsilon_c$, so $\beta\sigma_k \leq \beta\epsilon_c$ is the binding branch), i.e. $\kappa_\Delta = \beta$ works. This is the inequality-funnel analog of the criticality-step lemma of Conn et al. (2009, Ch. 10) realized by Algorithm_10_2.m. $\square$

Lemma 9 is the ingredient promised in Section 7.2: it ties Δ_k to the criticality σ_k . Together with L1 ($\|\nabla f - g_k\| \leq \kappa_{\text{eg}}\Delta_k$), it is what transfers model stationarity to true stationarity — the step that fully-linear models alone do not provide.

μ -iteration criticality.

Lemma 10 (μ -iterations are model-feasibility-critical). *Under the strengthened trigger (15), a μ -iteration ($s_k = 0$) has*

$$\chi_k^c = 0 \quad \text{and} \quad \pi_k^f \leq \omega_t(\theta_k).$$

Consequently (Lemma 11, contrapositive) a μ -iteration occurs only at points that are near-feasible relative to the radius, $\theta_k \leq (\kappa_{\text{eg}}^\theta \Delta_k / \kappa_m)^2 / 2$; in particular no μ -iteration has $\theta_k \geq \tau > 0$ with $\Delta_k < \kappa_m \sqrt{2\tau} / \kappa_{\text{eg}}^\theta$.

Proof. $s_k = 0$ forces $t_k = 0$ and $n_k = 0$. If switching (19) held, the tangential Cauchy decrease (22) is null only when $\pi_k^f = 0$, contradicting $\pi_k^f > \omega_t(\theta_k) \geq 0$; hence switching failed, $\pi_k^f \leq \omega_t(\theta_k)$. Then the trigger's second disjunct fired, so a normal step was computed; $n_k = 0$ makes the normal Cauchy bound (17) null, forcing $\chi_k^c = 0$. By Lemma 11, $0 = \chi_k^c \geq \kappa_m \sqrt{2\theta_k} - \kappa_{\text{eg}}^\theta \Delta_k$, i.e. $\sqrt{2\theta_k} \leq \kappa_{\text{eg}}^\theta \Delta_k / \kappa_m$, which is the stated near-feasibility bound. $\square$

The lemma is what lets L8 exclude μ -iterations (which shrink Δ) at infeasible points: a μ -iteration is model-feasibility-critical, and under extended MFCQ that means near-feasible relative to the radius. Its residual infeasibility, if any, is $O(\Delta_k^2)$ and handled by the funnel drive of T1.

8.3 Feasibility of limit points (Theorem T1)

The feasibility argument rests on the observation that MFCQ makes an infeasible point a *non-stationary* point of the infeasibility measure θ . We state the precise form used. For x with $\theta(x) > 0$ let the *positively-active set* be $\mathcal{A}^+(x) = \{i : c_i(x) \geq 0\}$ (constraints that are violated or exactly active); these are the ones contributing to θ .

Assumption 7 (Extended MFCQ on the working region). There is $\kappa_m > 0$ such that at every $x \in \mathcal{L}$ with $\theta(x) > 0$ there is a unit direction d with $\langle \nabla c_i(x), d \rangle \leq -\kappa_m$ for all $i \in \mathcal{A}^+(x)$.

Assumption 7 is the natural extension of MFCQ (Assumption 6) from the active set at feasible limit points to the positively-active set at infeasible points; at a feasible point ($\theta = 0$, $\mathcal{A}^+ = \mathcal{A}$) it is MFCQ. It is exactly the “no infeasible stationary points of θ ” hypothesis that the feasible-start scoping (Assumption 4) presumes: it guarantees the funnel can always make feasibility progress, so the method cannot stall at an infeasible point.

Lemma 11 (MFCQ excludes infeasible θ -stationarity). *Under Assumption 7, at every iteration the model feasibility criticality obeys*

$$\chi_k^c = \|\nabla \theta_k^m(x_k)\| \geq \kappa_m \sqrt{2\theta_k} - \kappa_{\text{eg}}^\theta \Delta_k. \quad (39)$$

In particular, for any $\varrho > 0$, if $\theta_k \geq \varrho$ and $\Delta_k \leq \bar{\Delta}(\varrho) := \kappa_m \sqrt{2\varrho} / (2\kappa_{\text{eg}}^\theta)$, then $\chi_k^c \geq \frac{1}{2} \kappa_m \sqrt{2\varrho} > 0$.

Proof. Because θ is smooth with $\nabla \theta(x_k) = J(x_k)^\top [c_k]_+ = \sum_{i:c_i(x_k)>0} c_i(x_k) \nabla c_i(x_k)$, the extended-MFCQ direction d (unit, $\langle \nabla c_i(x_k), d \rangle \leq -\kappa_m$ for $i \in \mathcal{A}^+(x_k) \supseteq \{i : c_i(x_k) > 0\}$) gives

$$\|\nabla \theta(x_k)\| \geq -\langle \nabla \theta(x_k), d \rangle = - \sum_{i:c_i(x_k)>0} c_i(x_k) \langle \nabla c_i(x_k), d \rangle \geq \kappa_m \sum_{i:c_i(x_k)>0} c_i(x_k) = \kappa_m \|[c_k]_+\|_1.$$

Since $\|[c_k]_+\|_1 \geq \|[c_k]_+\|_2 = \sqrt{2\theta_k}$, we get $\|\nabla \theta(x_k)\| \geq \kappa_m \sqrt{2\theta_k}$. The model gradient bound of L1, $\|\nabla \theta_k^m(x_k) - \nabla \theta(x_k)\| \leq \kappa_{\text{eg}}^\theta \Delta_k$, then yields (39). The “in particular” clause substitutes $\theta_k \geq \varrho$ and $\Delta_k \leq \bar{\Delta}(\varrho)$. $\square$

With the smooth measure this is a clean gradient bound — no subgradient or spurious-active-constraint bookkeeping. It is the exact model analog of the Gould–Toint feasibility bound $\|\nabla \theta\| \geq \kappa_m \sqrt{2\theta}$, and it is the pivot that excludes infeasible θ -stationary limit points in T1.

We give a *self-contained* proof that $\theta_k \rightarrow 0$, replacing the imported dichotomy of the previous draft. We follow the logical order of Gould and Toint (2010) — optimality progress is analyzed *before* feasibility, so no circular dependence on T2 arises — adapt every step to the derivative-free (f, c known only through fully-linear models, via L1) and smooth inequality ($\theta = \frac{1}{2} \|[c]_+\|^2$) setting, and use only the reproduced trust-region lemmas of Section 8.2 (nothing imported). Write $\kappa_{\text{dec}} = \kappa_{\text{tg}} \kappa_{\text{tc}}$ for the f -iteration decrease constant. On a successful f -iteration j , the ratio test gives $f(x_j) - f(x_{j+1}) \geq \eta_1 \text{pred}_j^f$; the f -domination test (25) gives $\text{pred}_j^f \geq \kappa_{\text{tg}} \delta_j^{f,t}$ where $\delta_j^{f,t} = m_f^j(x_j + n_j) - m_f^j(x_j + s_j)$ is the tangential objective decrease; and L3 bounds $\delta_j^{f,t}$ below. Chaining,

$$f(x_j) - f(x_{j+1}) \geq \eta_1 \kappa_{\text{tg}} \delta_j^{f,t} \geq \eta_1 \kappa_{\text{dec}} \pi_j^f \min\left(\frac{\pi_j^f}{1 + \kappa_H}, \Delta_j\right). \quad (40)$$

(Comparing pred_j^f to $\delta_j^{f,t}$ — not to pred_j^θ — is what makes this bound in terms of π_j^f valid; see (25).)

Lemma 12 (L6 — path length between criticality crossings). *Let $k_1 < k_2$ be successful iterations such that every successful iteration in $[k_1, k_2)$ is an f -iteration with $\pi_j^f \geq \varepsilon > 0$, and $f(x_{k_1}) - f(x_{k_2}) \leq \eta_1 \kappa_{\text{dec}} \varepsilon^2 / (2(1 + \kappa_H))$. Then*

$$\|x_{k_1} - x_{k_2}\| \leq \frac{1}{\eta_1 \kappa_{\text{dec}} \varepsilon} (f(x_{k_1}) - f(x_{k_2})). \quad (41)$$

Proof. For a successful f -iteration $j \in [k_1, k_2)$, (40) and $\pi_j^f \geq \varepsilon$ give $f(x_j) - f(x_{j+1}) \geq \eta_1 \kappa_{\text{dec}} \varepsilon \min(\varepsilon/(1 + \kappa_H), \Delta_j)$. Were the minimum ever $\varepsilon/(1 + \kappa_H)$, that single term would be $\geq \eta_1 \kappa_{\text{dec}} \varepsilon^2/(1 + \kappa_H) = 2[\eta_1 \kappa_{\text{dec}} \varepsilon^2/(2(1 + \kappa_H))] \geq 2(f(x_{k_1}) - f(x_{k_2}))$, exceeding the total decrease $\sum_j (f(x_j) - f(x_{j+1})) = f(x_{k_1}) - f(x_{k_2})$ (all terms nonnegative) — impossible. So the minimum is always Δ_j , whence $\|x_j - x_{j+1}\| = \|s_j\| \leq \Delta_j \leq (f(x_j) - f(x_{j+1})) / (\eta_1 \kappa_{\text{dec}} \varepsilon)$. On unsuccessful and μ -iterations x is unchanged. Summing over $j \in [k_1, k_2)$ and using the triangle inequality gives (41). $\square$

This is the derivative-free, inequality analog of [Gould and Toint \(2010, Lemma 3.20\)](#).

Lemma 13 (L7 — feasibility when finitely many successful c -iterations). *If $|\S_c| < \infty$ (Regime B), then $\theta_k \rightarrow 0$.*

Proof. Let k_0 be the last successful c -iteration. For $k > k_0$ every successful iteration is an f -iteration, so $\{f(x_k)\}_{k > k_0}$ is non-increasing (only f -iterations move x , and they decrease f) and bounded below (Proposition 1); it converges to some f_* . By (TR2) applied to these f -iterations (criticality π_k^f , merit f), $\liminf_k \pi_k^f = 0$.

Suppose $\theta_k \not\rightarrow 0$: there are $\varepsilon_0 > 0$ and infinitely many successful f -iterations $k > k_0$ with $\theta_k \geq \varepsilon_0$. On an f -iteration the switching test (19) held, so $\pi_k^f > \omega_t(\theta_k) \geq \omega_t(\varepsilon_0) > 0$ (ω_t non-decreasing, $\omega_t(t) > 0$ for $t > 0$). Fix ε with $0 < \varepsilon < \frac{1}{2}\omega_t(\varepsilon_0)$ and small enough that $\omega_t^{-1}(\varepsilon) < \varepsilon_0/2$ (possible as ω_t is strictly increasing near 0). Choose $k_1 > k_0$ with $\theta_{k_1} \geq \varepsilon_0$ and

$$f(x_{k_1}) - f_* \leq \eta_1 \kappa_{\text{dec}} \varepsilon \min\left(\frac{\varepsilon}{2(1 + \kappa_H)}, \frac{\varepsilon_0}{2L_\theta}\right), \quad (42)$$

where $L_\theta = \sup_{\mathcal{L}} \|\nabla \theta\| < \infty$ is the Lipschitz constant of the smooth θ ; possible since $f(x_k) \downarrow f_*$. Let $k_2 > k_1$ be the first successful iteration with $\pi_{k_2}^f < \varepsilon$ (it exists since $\liminf_k \pi_k^f = 0$ and $\pi_{k_1}^f \geq \omega_t(\varepsilon_0) > \varepsilon$); every successful iteration in $[k_1, k_2)$ is an f -iteration with $\pi_j^f \geq \varepsilon$. The first argument of (42) meets L6's hypothesis, so L6 gives $P := \sum_{j \in [k_1, k_2) \cap \S} \|x_j - x_{j+1}\| \leq (f(x_{k_1}) - f_*) / (\eta_1 \kappa_{\text{dec}} \varepsilon) \leq \varepsilon_0 / (2L_\theta)$. Each accepted step segment $[x_j, x_{j+1}]$ lies in $B(x_j, \Delta_j) \subseteq \mathcal{L}$, where $\|\nabla \theta\| \leq L_\theta$; the mean-value theorem on each segment and summation give

$$|\theta_{k_1} - \theta_{k_2}| \leq \sum_j |\theta(x_j) - \theta(x_{j+1})| \leq L_\theta \sum_j \|x_j - x_{j+1}\| = L_\theta P \leq \varepsilon_0/2,$$

so $\theta_{k_2} \geq \theta_{k_1} - \varepsilon_0/2 \geq \varepsilon_0/2$ (the polygonal path avoids assuming $\mathcal{L}$ convex). But k_2 is a successful f -iteration, so $\omega_t(\theta_{k_2}) < \pi_{k_2}^f < \varepsilon$, giving $\theta_{k_2} < \omega_t^{-1}(\varepsilon) < \varepsilon_0/2$ — a contradiction. Hence $\theta_k \rightarrow 0$. $\square$

This is the derivative-free, inequality analog of [Gould and Toint \(2010, Lemma 3.21\)](#); the inequality enters only through the 1-Lipschitz bound on $[\cdot]_+$.

Lemma 14 (L8 — feasibility when infinitely many successful c -iterations). *If $|\S_c| = \infty$ (Regime A), then $\theta_k \rightarrow 0$.*

Proof. By L4 the funnel converges, $\theta_k^{\max} \downarrow \theta_*^{\max} \geq 0$. Suppose $\theta_*^{\max} = \tau > 0$. On a successful c -iteration, L4-gap (36) and $\theta_k \leq \theta_k^{\max}$ telescope:

$$\sum_{k \in \S_c} \text{ared}_k^\theta \leq \sum_{k \in \S_c} (\theta_k^{\max} - \theta_{k+1}) \leq \frac{1}{1 - \kappa_{\text{cb}}} (\theta_0^{\max} - \tau) < \infty,$$

so $\text{ared}_k^\theta \rightarrow 0$, hence $\text{pred}_k^\theta \leq \text{ared}_k^\theta / \eta_1 \rightarrow 0$ on $\S_c$. Moreover, because $\kappa_{\text{ca}} \theta_k^{\max} \rightarrow \kappa_{\text{ca}} \tau < \tau$ while $\theta_{k+1}^{\max} \geq \tau$, for large $k \in \S_c$ the maximum in (29) is attained by its second argument, giving $\theta_{k+1} \geq (\tau - \kappa_{\text{cb}} \theta_k^{\max}) / (1 - \kappa_{\text{cb}}) \rightarrow \tau$; thus $\theta_k \geq \theta_{k+1} \geq \tau/2$ for all large $k \in \S_c$.

For all large $k \in \S_c$ we thus have $\theta_k \geq \tau/2 > 0$. A c -iteration computes a normal step, so the feasibility radius cap (14) is in force, and (Section 7.2) it makes cq-nonstat spurious-free:

$$\chi_k^c \geq \frac{\kappa_m \sqrt{2\theta_k}}{1 + \mu\kappa_{eg}^\theta} \geq \varsigma_\tau := \frac{\kappa_m \sqrt{\tau}}{1 + \mu\kappa_{eg}^\theta} > 0 \quad (k \in \S_c \text{ large}), \quad (43)$$

with no large-radius exception — the cap $\Delta_k \leq \mu\chi_k^c$ is exactly what prevents an over-large radius from masking the true criticality (this is where the smooth measure and the feasibility criticality step earn their keep). Every c -iteration carries a genuine normal step (Section 7.4), so the normal Cauchy bound (17) and feasibility-retention (21) ($\text{pred}_k^\theta \geq (1 - \kappa_{nt}) \times (17)$) give

$$\text{pred}_k^\theta \geq \frac{1 - \kappa_{nt}}{2} \chi_k^c \min\left(\frac{\chi_k^c}{1 + \beta_\theta}, \kappa_n \Delta_k\right) \quad \text{with } \chi_k^c \geq \varsigma_\tau. \quad (44)$$

Step 1 (summable radii on $\S_c$). Since $\chi_k^c \geq \varsigma_\tau$ is bounded below on large $\S_c$, (44) gives $\text{pred}_k^\theta \geq \frac{1 - \kappa_{nt}}{2} \varsigma_\tau \min(\varsigma_\tau/(1 + \beta_\theta), \kappa_n \Delta_k)$. At most finitely many $k \in \S_c$ have $\Delta_k \geq \varsigma_\tau/(\kappa_n(1 + \beta_\theta))$ (each such has pred_k^θ bounded below, but $\text{pred}_k^\theta \rightarrow 0$); for the rest the min is $\kappa_n \Delta_k$, so $\text{pred}_k^\theta \geq c_2 \Delta_k$ with $c_2 := \frac{(1 - \kappa_{nt})\kappa_n \varsigma_\tau}{2}$. With the telescoping bound, $\sum_{k \in \S_c} \Delta_k \leq c_2^{-1} \sum_{k \in \S_c} \text{pred}_k^\theta + O(1) < \infty$. (No trust-region radius-bookkeeping is imported here: the cap (14) makes χ_k^c genuinely bounded below, so the earlier “large-radius spurious criticality” case cannot arise.) (*)

Step 2 (finitely many band up-crossings). Let $L_f = \sup_{\mathcal{L}} \|\nabla f\|$ and $L_\theta = \sup_{\mathcal{L}} \|\nabla \theta\|$ (finite, Assumptions 2, 3), $\pi_B := \omega_t(\tau/4) > 0$, and consider the band $\{x : \tau/4 \leq \theta(x) \leq \tau/2\}$. An up-crossing is an ascent from $\theta \leq \tau/4$ to $\theta \geq \tau/2$. Only f -iterations raise θ (a c -iteration reduces it; μ /unsuccessful leave x fixed), and a rising f -iteration in the band has, by switching (19), $\pi_j^f > \omega_t(\theta_j) \geq \pi_B$; hence by (40), when $\Delta_j \leq \pi_B/(1 + \kappa_H)$,

$$f(x_j) - f(x_{j+1}) \geq \eta_1 \kappa_{\text{dec}} \pi_B \Delta_j \geq \frac{\eta_1 \kappa_{\text{dec}} \pi_B}{L_\theta} (\theta_{j+1} - \theta_j)$$

(using $\theta_{j+1} - \theta_j \leq L_\theta \|s_j\| \leq L_\theta \Delta_j$), and $f(x_j) - f(x_{j+1}) \geq \eta_1 \kappa_{\text{dec}} \pi_B^2/(1 + \kappa_H)$ otherwise. Summing the rising f -iterations of one up-crossing (whose θ -increases total $\geq \tau/4$) yields an objective decrease $\geq \Phi := \min\left(\frac{\eta_1 \kappa_{\text{dec}} \pi_B}{L_\theta} \frac{\tau}{4}, \frac{\eta_1 \kappa_{\text{dec}} \pi_B^2}{1 + \kappa_H}\right) > 0$ per up-crossing. Meanwhile f increases only on successful c -iterations, each by $\leq L_f \Delta_k$, so the total increase is $\leq L_f \sum_{k \in \S_c} \Delta_k < \infty$ by (*). Since $f \geq f_{\text{low}}$ (Proposition 1),

$$(\#\text{up-crossings}) \cdot \Phi \leq f(x_0) - f_{\text{low}} + L_f \sum_{k \in \S_c} \Delta_k < \infty,$$

so there are finitely many up-crossings.

Step 3 (contradiction). After the last up-crossing, either (I) $\theta_k \geq \tau/4$ for all large k , or (II) $\theta_k < \tau/2$ for all large k .

In case (II) only finitely many k have $\theta_k \geq \tau/2$; but every large $k \in \S_c$ has $\theta_k \geq \tau/2$, so $|\S_c| < \infty$ — contradicting Regime A.

In case (I) all large iterations lie in the pure infeasible region $\{\theta \geq \tau/4\}$ (no near-feasible excursions interleave). There cq-nonstat (39) gives, for $\Delta_k \leq \varrho_0 := \kappa_m \sqrt{\tau/2}/(2\kappa_{eg}^\theta)$, $\chi_k^c \geq \frac{1}{2} \kappa_m \sqrt{\tau/2} =: \varsigma_0 > 0$; so no μ -iteration occurs (Lemma 10 needs $\chi_k^c = 0$), and Δ can shrink only on unsuccessful iterations. Apply the reproduced accuracy lemma 5: a c -iteration (merit θ , $\xi = \chi_k^c \geq \varsigma_0$, decrease (44)) and an f -iteration (merit f , $\xi = \pi_k^f > \pi_B$ by switching, decrease (40)) are both successful once $\Delta_k \leq \Delta_3 := \min(\varrho_0, \bar{\Delta}(\varsigma_0), \bar{\Delta}(\pi_B))$ (the accuracy thresholds of Lemma 5 for the two merits). Hence no shrinkage occurs at $\Delta_k \leq \Delta_3$, so by Lemma 6 $\Delta_k \geq \gamma \Delta_3 =: \Delta_{\text{lb}} > 0$ for all large k . But this contradicts (*) ($\sum_{k \in \S_c} \Delta_k < \infty$, with $|\S_c| = \infty$, forces $\Delta_k \rightarrow 0$ along $\S_c$).

Both cases are impossible, so $\tau = 0$: $\theta_k^{\text{max}} \rightarrow 0$ and $\theta_k \leq \theta_k^{\text{max}} \rightarrow 0$. $\square$

Theorem 1 (T1 — feasibility). *Under Assumptions 1–6 and 7, $\theta_k \rightarrow 0$; hence $\theta_*^{\max} = 0$ and every limit point x_* of $\{x_k\}$ is feasible, $c(x_*) \leq 0$.*

Proof. Either $|\S_c| < \infty$ (L7) or $|\S_c| = \infty$ (L8); both give $\theta_k \rightarrow 0$. By continuity of c , $c(x_*) = \lim c(x_k) \leq 0$ at every limit point. $\square$

Remark 1 (The feasibility theorem is self-contained). *Theorem 1 is proved in full. Every trust-region ingredient it uses — the accuracy-implies-success lemma 5, the radius lower bound 6, and the liminf-criticality 7 (invoked in Regime B) — is reproduced in Section 8.2; nothing is imported. The smooth measure $\theta = \frac{1}{2}\|c\|_+^2$ is what makes this clean: it turns feasibility into smooth trust-region minimization with the gradient criticality $\chi^c = \|\nabla\theta^m\|$, and the feasibility radius cap (14) removes the large-radius “spurious criticality” obstruction outright (the former residual TD1’ is gone — see Step 1 of L8). The genuinely new content is the extended-MFCQ pivot (Lemma 11, now a one-line gradient bound $\chi^c \geq \kappa_m\sqrt{2\theta} - \kappa_{\text{eg}}^\theta\Delta$), the funnel-inequality interaction, and the band-crossing argument that tames the interleaving of infeasible and near-feasible iterations.*

The pivot throughout is Lemma 11: extended MFCQ forces $\chi_k^c > 0$ at infeasible points, ruling out the “infeasible stall” a pure funnel-relative trigger would permit. This is where the feasible-reachable-start scoping (Assumption 4) pays off: it is consistent with the absence of infeasible θ -stationary traps that Assumption 7 formalizes.

8.4 Optimality: a KKT limit point (Theorem T2)

With feasibility secured, the optimality argument applies the reproduced trust-region lemmas of Section 8.2 to the f -iterations, which dominate once $\theta_k \rightarrow 0$.

Lemma 15 (Eventually f -iterations near a non-KKT accumulation). *If $\pi_k^f \geq \pi_* > 0$ along a subsequence, then along that subsequence the switching condition (19) holds for all large k and each such iteration is an f -iteration.*

Proof. By T1, $\theta_k \rightarrow 0$. Since $\pi_k^f \geq \pi_*$ and $\omega_t(\theta_k) \rightarrow \omega_t(0) = 0$, the switching test (19) $\pi_k^f > \omega_t(\theta_k)$ holds for all large k in the subsequence, so a tangential step is taken and, by L3,

$$\delta_k^{f,t} \geq \kappa_{\text{tc}} \pi_* \min(\pi_*/(1 + \kappa_H), \Delta_k) > 0.$$

We must show the f -domination test (25), $\text{pred}_k^f \geq \kappa_{\text{tg}} \delta_k^{f,t}$, so the iteration is classified f . Writing $\text{pred}_k^f = \delta_k^{f,t} + [m_f^k(x_k) - m_f^k(x_k + n_k)]$, the normal-step objective change is bounded using $\|n_k\| \leq \kappa_{\text{nn}}\sqrt{2\theta_k}$ (18) and $\|g_k\|, \|H_k\|$ bounded (Proposition 1):

$$|m_f^k(x_k) - m_f^k(x_k + n_k)| \leq \|g_k\| \|n_k\| + \frac{1}{2}\|H_k\| \|n_k\|^2 \leq C\sqrt{\theta_k}$$

for a constant C (as $\theta_k \rightarrow 0$, $\sqrt{\theta_k}$ dominates θ_k). Near a non-critical accumulation the criticality step (L5) keeps $\Delta_k \geq \kappa_\Delta \sigma_k \geq \kappa_\Delta \pi_*$ bounded below, so $\delta_k^{f,t} \geq \kappa_{\text{tc}} \pi_* \min(\pi_*/(1 + \kappa_H), \kappa_\Delta \pi_*) =: \delta_* > 0$ is bounded below, while $C\sqrt{\theta_k} \rightarrow 0$ (by T1). Hence for all large k , $C\sqrt{\theta_k} \leq (1 - \kappa_{\text{tg}})\delta_* \leq (1 - \kappa_{\text{tg}})\delta_k^{f,t}$, giving $\text{pred}_k^f \geq \delta_k^{f,t} - C\sqrt{\theta_k} \geq \kappa_{\text{tg}} \delta_k^{f,t}$. Thus (25) holds and the iteration is an f -iteration. $\square$

Theorem 2 (T2 — first-order optimality / KKT). *Under Assumptions 1–6 and 7,*

$$\liminf_{k \rightarrow \infty} \pi_k^f = 0.$$

Consequently there is a limit point x_ of $\{x_k\}$ that is feasible ($c(x_*) \leq 0$, by T1) and satisfies the KKT conditions of (1): there exists $\lambda \geq 0$ with $\nabla f(x_*) + \sum_i \lambda_i \nabla c_i(x_*) = 0$ and $\lambda_i c_i(x_*) = 0$. Under the criticality step (Lemma 9) the stronger $\lim_{k \rightarrow \infty} \pi_k^f = 0$ holds, so every feasible limit point is KKT.*

Proof. $\liminf \pi_k^f = 0$. Suppose not: $\pi_k^f \geq \pi_* > 0$ for all large k . By Lemma 15 all large iterations are f -iterations, so f is non-increasing over the tail (only f -iterations move x , decreasing f ; the others fix it) and bounded below (Proposition 1); and L3 supplies the fraction-of-Cauchy decrease (37) for the merit f with criticality π^f . The reproduced Lemma 7 (TR2), applied with $\varphi = f$, $\xi = \pi^f$, then gives $\liminf_k \pi_k^f = 0$ — contradicting $\pi_k^f \geq \pi_*$.

KKT limit point. Take a subsequence with $\pi_k^f \rightarrow 0$ and (by boundedness of $\mathcal{L}$) $x_k \rightarrow x_*$. By T1, $\theta(x_*) = 0$, so x_* is feasible. On this subsequence the criticality step drives $\Delta_k \rightarrow 0$ (L5, since $\sigma_k \rightarrow 0$), so L1 gives $g_k \rightarrow \nabla f(x_*)$ and $J_k \rightarrow \nabla c(x_*)$.

Active-set identification (shrinking ε_k). The model active set uses the *shrinking* tolerance $\varepsilon_k = \kappa_\varepsilon \sigma_k \rightarrow 0$ (Section 7.1), so $\mathcal{A}_k(\varepsilon_k) = \{i : c_i(x_k) \geq -\varepsilon_k\}$. For $i \notin \mathcal{A}(x_*)$ we have $c_i(x_*) < 0$; by continuity $c_i(x_k) \rightarrow c_i(x_*) < 0$, so for large k , $c_i(x_k) < -\varepsilon_k$ and $i \notin \mathcal{A}_k(\varepsilon_k)$. Conversely each $i \in \mathcal{A}(x_*)$ ($c_i(x_*) = 0$) satisfies $c_i(x_k) \geq -\varepsilon_k$ for large k . Hence $\mathcal{A}_k(\varepsilon_k) = \mathcal{A}(x_*)$ for all large k in the subsequence — exact identification, which a *fixed* ε would not deliver.

Passing to the limit. The model KKT residual $\pi_k^f \rightarrow 0$ furnishes multipliers $\lambda^k \geq 0$ supported on $\mathcal{A}_k(\varepsilon_k) = \mathcal{A}(x_*)$ with $g_k + (J_k^A)^\top \lambda^k \rightarrow 0$. Since $\mathcal{A}_k(\varepsilon_k) = \mathcal{A}(x_*)$ and $J_k^A \rightarrow \nabla c_{\mathcal{A}}(x_*)$, MFCQ (Assumption 6) at x_* — linear independence of a strictly-feasible direction, equivalently boundedness of the multiplier set — bounds $\{\lambda^k\}$, so it has a limit point $\lambda \geq 0$ supported on $\mathcal{A}(x_*)$. Passing to the limit in $g_k + (J_k^A)^\top \lambda^k \rightarrow 0$ gives $\nabla f(x_*) + \nabla c(x_*)^\top \lambda = 0$; complementarity $\lambda_i c_i(x_*) = 0$ holds because $\lambda_i = 0$ for $i \notin \mathcal{A}(x_*)$ (multipliers supported on $\mathcal{A}(x_*)$) and $c_i(x_*) = 0$ for $i \in \mathcal{A}(x_*)$. Thus x_* is a KKT point.

Strengthening to $\lim \pi_k^f = 0$. Once $\theta_k \rightarrow 0$ the iterations are eventually f -iterations wherever π_k^f is bounded below (Lemma 15), so the reproduced Lemma 8 (TR-lim) applies with $\varphi = f$, $\xi = \pi^f$, giving $\lim_k \pi_k^f = 0$; hence *every* limit point is a KKT point. $\square$

8.5 Limiting-case validations

We check that the analysis reduces correctly in the four regimes named in the plan; these are the sanity constraints on the proof.

(a) $p = 0$ (unconstrained) $\Rightarrow$ CSV Ch. 10/11. With no constraints, $\theta \equiv 0$, $\chi_k^c \equiv 0$, $n_k \equiv 0$, $\theta_k^{\max}$ is irrelevant, and every iteration is an f -iteration with $\pi_k^f = \|g_k\|$ and $\mathcal{A}_k = \emptyset$. Algorithm 1 becomes `Algorithm_11_1.m`, L3 becomes the unconstrained Cauchy decrease, L5 becomes the CSV criticality step, T1 is vacuous, and T2 reduces to $\liminf \|g_k\| = 0$ (resp. $\lim \|g_k\| = 0$), i.e. CSV Ch. 10/11.

(b) Equalities $\Rightarrow$ Sampaio–Toint (2015). An equality $c^{\text{eq}}(x) = 0$ is $\{c^{\text{eq}} \leq 0, -c^{\text{eq}} \leq 0\}$; then $[c]_+$ recovers $|c^{\text{eq}}|$ componentwise, so $\theta = \frac{1}{2} \|[c]_+\|^2 = \frac{1}{2} \|c^{\text{eq}}\|^2$ with $\nabla \theta = (J^{\text{eq}})^\top c^{\text{eq}}$ and feasibility criticality $\chi^c = \|(J^{\text{eq}})^\top c^{\text{eq}}\|$. This is *exactly* the squared infeasibility measure $\frac{1}{2} \|c\|^2$ and criticality $\|J^\top c\|$ of Sampaio and Toint (2015): having adopted the smooth measure, the correspondence is now a genuine specialization (not merely structural), with L2/L4/T1 specializing to their feasibility lemmas and T2 to their first-order result. The only differences are the inequality one-sidedness ($[\cdot]_+$) and the derivative-free model layer (L1).

(c) MFCQ failure $\Rightarrow$ identifies the breaking lemma. If Assumption 7 fails at some infeasible $\bar{x} \in \mathcal{L}$ (no direction strictly reduces all positively-active constraints), then Lemma 11 fails: χ_k^c may vanish with $\theta_k > 0$, so L2's right-hand side is 0 and T1's feasibility drive stalls — the algorithm may converge to an *infeasible* stationary point of θ . This is precisely the dropped dichotomy branch;

it localizes the indispensability of MFCQ to Lemma 11 $\rightarrow$ T1. If instead MFCQ (Assumption 6) fails only at a *feasible* limit point, T1 still holds but the multiplier set in T2 may be unbounded, breaking the KKT conclusion (only Fritz–John stationarity survives).

(d) Feasible-iterate regime $\Rightarrow$ NOWPAC relation. If the acceptance rule is tightened to admit only $\theta(x_k + s_k) = 0$ (equivalently $\theta_k^{\max} \equiv 0$), then no relaxable excursion is ever taken, the normal step keeps iterates feasible, and Algorithm 1 runs entirely inside $\mathcal{F}$. This is the feasible-iterate regime of NOWPAC (Augustin and Marzouk, 2017; Menhorn et al., 2022); our analysis then does not *use* relaxability, and T1 is automatic ($\theta_k \equiv 0$). The contrast is that our default ($\theta_k^{\max} > 0$) *exploits* relaxability to sample and step through infeasible points, which is unavailable to a feasible-iterate method.

8.6 Summary of the argument

The charter question — “do fully-linear models of f and c on every iteration suffice for convergence to a KKT point?” — is answered: *necessary, but only in concert with two further mechanisms*. L1 shows fully-linearity of f and c (and the inherited accuracy of θ) controls all four ratio tests; L5 (the criticality step) ties Δ_k to criticality so model stationarity transfers to true stationarity; and the feasibility theorem 1 — proved self-contained via the path-length lemma 12, the two regime lemmas 13–14, and the extended-MFCQ Lemma 11 — drives $\theta_k \rightarrow 0$. Remove any one and a limiting-case check in Section 8.5 breaks. The two theorems T1 (feasibility) and T2 (KKT under MFCQ) are the deliverable D3. *The analysis is now self-contained*: the single-region trust-region facts it needs (Lemmas 5–8) are reproduced in Section 8.2 from first principles, and the smooth measure $\theta = \frac{1}{2}\|[c]_+\|^2$ with the feasibility radius cap (14) removes the large-radius obstruction that the earlier unsquared draft could only cite.

Status

Milestones M1–M4 are complete (problem/notation, algorithm, convergence analysis, and an adversarial self-review with the D5 code-bridge appendix), and both prior technical-debt items are now closed:

TD1 (self-contained feasibility) — resolved. The feasibility theorem 1 ($\theta_k \rightarrow 0$) is proved directly in both regimes (Section 8.3): Lemma 12 (path length), 13 (finitely many successful c -iterations), 14 (infinitely many), in the Gould–Toint logical order (optimality before feasibility). The *interleaving* of infeasible and near-feasible iterations is handled self-contained by a summable-radii bound and a *finitely-many-band-crossings* argument; the imported trust-funnel dichotomy is gone.

TD2 (single-region trust-region machinery) — resolved. Two design changes requested and made: (i) the *smooth* infeasibility measure $\theta = \frac{1}{2}\|[c]_+\|^2$ (Section 3), which is C^1 with $\nabla\theta = J^\top[c]_+$, turns feasibility into smooth trust-region minimization with the gradient criticality $\chi^c = \|\nabla\theta^m\|$ and makes the MFCQ pivot (Lemma 11) a one-line bound $\chi^c \geq \kappa_m\sqrt{2\theta} - \kappa_{\text{eg}}^\theta\Delta$; and (ii) the standard single-region trust-region results are now *reproduced from first principles* in Section 8.2 (Lemmas 5 accuracy $\Rightarrow$ success, 6 radius lower bound, 7 liminf criticality, 8 full criticality limit), applied to both merits f and θ . The feasibility radius cap (14) removes the former large-radius “spurious criticality” obstruction (the earlier TD1’) outright.

The convergence analysis is now self-contained, resting only on the stated assumptions and results proved within: T1 (feasibility, $\theta_k \rightarrow 0$) and T2 ($\lim_k \pi_k^f = 0$, every feasible limit point

KKT under MFCQ). The constrained/relaxable novelty — extended MFCQ excluding infeasible θ -stationarity, the funnel–inequality interaction, and the band-crossing feasibility argument — is proved in full; the reproduced single-region lemmas cite [Conn et al. \(2000, Ch. 6.4\)](#) only as the source whose arguments they restate.

A Code-bridge appendix (D5)

This appendix maps each component of Algorithm 1 (Section 7) to an existing routine in this repository, so the later implementation phase is a translation rather than a fresh build. **No code is written in this theory phase.** Paths are relative to the repository root; all listed files were confirmed present at the time of writing. Each row also flags whether the block is *reusable as-is*, needs *adaptation*, or is a genuine *extension point* (new code the later phase must add).

Algorithm 1 component	Existing building block	Status
Step 1 fully-linear models of $F = (f, c_1, \dots, c_p)$; g_k, J_k , model Hessians; <code>valid</code> flag (Assumption 5)	<code>pounders/py/formquad.py</code> — vector-valued minimum-Frobenius-norm models, returns <code>G</code> , <code>H</code> , and <code>valid</code> ; monomial basis in <code>phi2eval.py</code>	reuse as-is
Λ -poised geometry management (uniform constants of L1/L5)	<code>DFO_quad_TR/Algorithm_6_4.m</code> , <code>Model_Improvement_Step.m</code> , <code>Maximize_Lagrange_polynomial.m</code> , <code>Test_fully_quadratic.m</code> ; geometry rays via <code>pounders/py/bmpts.py</code> + <code>boxline.py</code>	reuse as-is
Infeasibility measure $\theta = \ [c]_+\ $ and its model θ_k^m (5)–(7)	compose from the vector model m_c^k (from <code>formquad</code>); the composition pattern mirrors <code>pounders/py/general_h_funs.py</code> <code>combine_*</code> (e.g. <code>combine_least_squares</code>)	adapt
Two criticality measures χ_k^c (8), π_k^f (9)	new: convex-composite steepest-descent slope for χ_k^c ; projected-gradient/KKT residual for π_k^f over J_k^A	extension point
Step 1 criticality loop (L5), Δ tie-in (13)	<code>DFO_quad_TR/Algorithm_10_2.m</code> — structure and $\Delta = \min(\max(\Delta, \beta\sigma), \Delta_{\text{in}})$ carried over verbatim; only $\sigma_k = \max(\chi_k^c, \pi_k^f)$ replaces the scalar criticality	adapt
Step 2a normal step n_k : $\min \ [c_k + J_k n]_+\ $ s.t. $\ n\ \leq \kappa_n \Delta$ (16)	convex QP/LP in (c_k, J_k) ; box-constrained core solvers <code>pounders/py/bqmin.py</code> (<code>spsolver=1</code>) or external <code>minq</code> submodule (<code>spsolver=2</code> , Neumaier <code>minq5</code>); needs an inequality/violation layer on top	extension point
Step 2b tangential step t_k : reduce m_f^k on $\{J_k^A t \leq 0\} \cap B$ (20)	quadratic-model minimization over box: <code>bqmin.py/minqsw</code> ; the null-space/active-cone restriction $J_k^A t \leq 0$ is added on top	adapt
Step 3 ratios ρ_k^f, ρ_k^c ; f -/ c -iteration classification (26)–(28)	pattern of <code>DFO_quad_TR/Algorithm_11_1.m</code> Step 3 (<code>rho</code> test); add the θ -ratio ρ_k^c and the switching-based classification	adapt
Step 4b funnel update $\theta_{k+1}^{\max}$ (29)	new scalar bookkeeping (Gould–Toint rule)	extension point
Step 5 radius update (30)	<code>DFO_quad_TR/Algorithm_11_1.m</code> Step 5 — carried over verbatim	reuse as-is
Blackbox eval of F (one call returns f and all c_i)	driver pattern in <code>Algorithm_11_1.m</code> ; input checks <code>pounders/py/checkinputss.py</code>	reuse as-is

Deliberate non-reuse (penalty). The repository contains a penalty-style composition, `h_max_plus_quadratic` in `manifold_sampling/py/general_nonsmooth_h_funs.py`. Consistent with the design decision recorded in Section 6 (trust-funnel, *not* penalty), Algorithm 1 does **not** use it: infeasibility is controlled by the funnel bound $\theta_k^{\max}$, never folded into the objective through a penalty parameter. It is listed here only to mark it as explicitly out of scope.

Manifold/active-set machinery for the nonsmooth θ . The nonsmoothness of $\theta = \|[c]_+\|$ at active/violated boundaries (Section 3) is handled in the analysis by the convex-composite argument. For a robust implementation of χ_k^c and the model active set $\mathcal{A}^+$, the manifold-sampling machinery (`manifold_sampling/py/build_p_models.py`, `h_pw_maximum` and relatives in `general_nonsmooth_h_funs.py`) is the natural tool for identifying the active manifold of $[\cdot]_+$ — reused for

its *geometry/active-set* identification, not for any penalty semantics.

Summary of extension points (the new code the next phase must write).

1. the two criticality measures χ_k^c, π_k^f ;
2. the linearized-violation normal-step subproblem (16) on top of the box-QP solvers;
3. the active-cone restriction for the tangential subproblem (20);
4. the funnel bookkeeping (bound $\theta_k^{\max}$, update, and the f -/ c -iteration switch).

Everything else — model building, poisedness/geometry, the criticality-loop skeleton, ratio tests, and the radius update — is already present and reused.

References

- Florian Augustin and Youssef M. Marzouk. NOWPAC: A provably convergent derivative-free nonlinear optimizer with path-augmented constraints. *arXiv preprint arXiv:1403.1931*, 2017. Original NOWPAC; feasible-iterate inner-boundary path.
- E. G. Birgin, N. Krejić, and J. M. Martínez. Inexact restoration for derivative-free expensive function minimization and applications. *Journal of Computational and Applied Mathematics*, 2021. File: articles/8025.pdf.
- Andrew R. Conn, Nicholas I. M. Gould, and Philippe L. Toint. *Trust-Region Methods*. MPS-SIAM Series on Optimization. SIAM, Philadelphia, 2000. doi: 10.1137/1.9780898719857. Single-region trust-region convergence theory (criticality step, radius lower bound, liminf/lim criticality).
- Andrew R. Conn, Katya Scheinberg, and Luís N. Vicente. *Introduction to Derivative-Free Optimization*. MOS-SIAM Series on Optimization. SIAM, Philadelphia, 2009. doi: 10.1137/1.9780898718768. File: articles/Conn, Scheinberg, Vicente - 2009 ...pdf.
- N. I. M. Gould and Ph. L. Toint. Nonlinear programming without a penalty function or a filter. *Mathematical Programming, Series A*, 122(1):155–196, 2010. doi: 10.1007/s10107-008-0244-7. File: articles/s10107-008-0244-7.pdf.
- Jeffrey Larson, Matt Menickelly, and Stefan M. Wild. Derivative-free optimization methods. *Acta Numerica*, 28:287–404, 2019. doi: 10.1017/S0962492919000060. File: articles/1904.11585v2.pdf.
- Sébastien Le Digabel and Stefan M. Wild. A taxonomy of constraints in simulation-based optimization. Technical Report 1505.07881, arXiv, 2015. Defines relaxable / quantifiable / simulation constraints (QRSK taxonomy).
- Friedrich Menhorn, Florian Augustin, Hans-Joachim Bungartz, and Youssef M. Marzouk. A trust-region method for derivative-free nonlinear constrained stochastic optimization. Technical Report 1703.04156v2, arXiv, 2022. (S)NOWPAC; extends NOWPAC. File: articles/1703.04156v2.pdf.
- Ph. R. Sampaio and Ph. L. Toint. A derivative-free trust-funnel method for equality-constrained nonlinear optimization. *Computational Optimization and Applications*, 61(1):25–49, 2015. doi: 10.1007/s10589-014-9715-3. File: articles/s10589-014-9715-3.pdf.
- Ph. R. Sampaio and Ph. L. Toint. Numerical experience with a derivative-free trust-funnel method for nonlinear optimization problems with general nonlinear constraints. *Optimization Methods and Software*, 31(3):511–534, 2016. doi: 10.1080/10556788.2015.1135919.